Perspective Coding Theory

— Holographic, Rateless, and Covert Channel Coding from Harmonic Prefix Vectors

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Abstract. This paper proposes and rigorously analyzes a class of deterministic channel codes: perspective codes. A message $b \in \{0,1\}^n$ is encoded into a signed harmonic prefix vector $E(b) \in \mathbb{R}^{n+1}$, $E(b)_k = \pm\frac{1/k}{H_{n+1}}$ (the signs carry the message, H denotes the harmonic numbers, and the first coordinate is fixed positive as an anchor). The code is completely characterized by four axioms (orthogonal branching, normalization conservation, perfect memory, and projective readout). This paper proves that the encoding is uniquely decodable and supports two decoding modes, direct-read and peeling (Theorem 4.1); that any coordinate subset directly yields the corresponding substring, with access complexity independent of n (Theorem 4.2); that decoding is strictly immune to global scaling and to arbitrary per-coordinate positive distortion (Theorems 5.1 and 5.2), and remains complete in projective space (Theorem 5.3); that the prefix property makes the code naturally rateless—the encoding of the first m bits is exactly the restriction of the full-length encoding to the first $m+1$ coordinates, multiplied by H_{m+1}/H_{n+1} (Theorem 6.1); that the coordinate levels $1/(kH_{n+1})$ form a logarithmically graded unequal error protection profile with dynamic range $n+1$ (Theorem 6.2); and that the modulus readout carries no message information whatsoever, $I(|E|;b) = 0$ exactly (Theorem 6.3); the amplitude layer and the sign layer are strictly orthogonal and informationally isolated, so the amplitude layer can carry an independent, conventionally coded stream unrelated to the sign stream, while the sign layer is superimposed at zero marginal energy cost (Theorem 9.9). Under additive white Gaussian noise, with $\sigma_0 = 1/((n+1)H_{n+1})$ (the weakest coordinate level) as the noise unit: the sign-constrained capacity is $C_{\mathrm{BPSK}} \approx 0.838n$, the hard-decision rate $\approx 0.775n$, and the rate after 20% erasure $\approx 0.688n$ (§9.3); the total capacity has the gamma closed form $C_{\mathrm{Sh}}(n) = \frac{1}{2}\log_2\prod_{k=1}^{n+1}(1+((n+1)/k)^2) = \frac{\pi+2\ln 2}{4\ln 2}\,n + O(\log n) \approx 1.6331n$ (Theorem 9.2); the energy accounting is settled rigorously: the reliable precision of the sign layer improves at least exponentially with the SNR and the reliable digits per joule are flat across the ladder (Proposition 9.5), so the $+10\,\mathrm{dB}$ per decade growth of E_b/N_0 is the price of ladder depth (equal-target protection down to the weakest coordinate), not an efficiency defect; the anchor share of the energy tends to $6/\pi^2 \approx 60.8\%$ (§9.4); on fading channels the multi-ledger identity $I(r;b,G) = I(r;b) + I(r;G\,|\,b) = I(r;G) + I(r;b\,|\,G)$ closes term by term exactly, verified by independent numerical integration (§9.6); freeing the compression ratio λ_r as a second message parameter yields a dual-channel code independent of the sign channel (Theorem 9.7); and switching to per-coordinate $2Q_k$-PAM attains 94.9–95.0% C_{Sh} with a residual gap of 0.082 bits per layer, at the price of losing covertness and distortion immunity (§9.8). Every assertion is double-checked by exact rational arithmetic and by Monte Carlo simulation ($\geq 10^4$ trials per configuration, 2×10^4 in most tables); Appendix A gives five algorithms, Appendix M collects all mapping tables, and Appendix B provides twelve data tables.

Keywords: channel coding; rateless codes; unequal error protection; physical-layer security; holographic storage; harmonic numbers; AWGN capacity; fading channels

Contents

§1 Introduction

Classical channel coding maps a message to a codeword whose coordinates are typically equal in status: deleting any segment is equally costly, and amplifying any segment changes nothing about the meaning. This paper studies a code whose coordinates are inherently unequal: the level of the k-th coordinate follows the harmonic proportion $1/k$, so that the signal-to-noise ratios of the strongest and weakest coordinates differ by a factor of $(n+1)^2$. Superficially this looks wasteful—the weak coordinates are almost buried in noise. The analysis shows that this inequality alone yields four properties that classical codes obtain only at extra cost: (1) **Ratelessness**: every prefix is a legal codeword, and the prefix codeword encodes exactly the prefix message (Theorem 6.1)—this is the property of fountain codes (Luby's LT codes [1]), but here no random degree distribution is needed and the structure is deterministic; (2) **Logarithmic unequal error protection (UEP)**: the error probability degrades smoothly along the coordinates from machine precision to $Q(1)$ (Theorem 6.2), so placing important bits earlier gives graded protection—a free realization, inside a single-user code, of the degraded broadcast channel capacity region [2]; (3) **Amplitude covertness**: the message lives entirely in the signs (phases), and an envelope observer obtains exactly zero message information (Theorem 6.3)—information-theoretic physical-layer security depending on no computational assumption; (4) **Holographic addressing**: a coordinate subset directly yields the corresponding message substring, without consulting the whole (Theorem 4.2).

The costs are equally explicit, and this paper does not evade them: the rate of the sign layer is $n/(n+1)$ (n bits over $n+1$ coordinates, asymptotically 1 bit per coordinate); its sign-constrained achievable rate is about half of the Shannon capacity of the same channel ($\eta(\sigma_0) = 0.513$, Remark 9.2.1); and the SNR of the weakest coordinate decays logarithmically in n; extending equal-target protection along the UEP ladder down to the weakest coordinate carries an energy cost growing at $+10$ dB per decade in n (§9.4). But this paper proves rigorously that this is not an efficiency defect: the reliable precision of the sign layer improves at least exponentially with the SNR—every additive increase of $2\ln 10 \approx 4.6052$ in the SNR delivers one more reliable decimal digit on every coordinate (Proposition 9.5)—and the reliable digits delivered per coordinate are proportional to its energy with a universal constant (Remark 9.5.1). More fundamentally, the amplitude layer and the sign layer are strictly orthogonal and informationally isolated (Theorem 9.9): the amplitude layer may use virtually any conventional coding to carry an independent stream—the two layers need not carry the same or similar information—while the sign layer is superimposed at zero marginal energy cost and keeps all of its structural advantages. The $2Q$-PAM modification (§9.8) is exactly an instance of this isolation theorem, reaching 94.9–95.0% of the Shannon capacity at the price of covertness and distortion immunity alone (a consequence of the amplitude layer starting to carry information). The capacity analysis of §9 settles every account: which properties are intrinsic to the structure and which are the price of design choices.

Mathematically, the perspective code is the direct engineering of the harmonic transition triangle of the companion paper *Fractal Structure and Logical-Operational Generation of Primes* (hereafter "the companion paper"): the generation rule $P^{(r+1)} = \lambda_r P^{(r)} \oplus (1-\lambda_r)\, e_{r+1}$ (with $\lambda_r = H_r/H_{r+1}$) there becomes the encoding recursion here, the perfect-memory law there becomes the peeling decoder here, and the p-adic new-letter events there become the prime frame-synchronization markers here (Theorem 8.3). The paper is self-contained: every fact about harmonic numbers that it needs is re-established and proved in §2–§3.

Relation to the literature: the AWGN capacity and its per-coordinate allocation belong to the standard theory of Shannon [3][4] and Cover–Thomas [5]; the idea of a per-subchannel SNR ladder is of the same family as bit loading for DSL discrete multitone modulation [6]; the shaping loss $\frac{1}{2}\log_2(\pi e/6)$ is due to Forney–Wei [7]; ratelessness is compared with LT codes [1]; and the difference from the polar-code route [8] is made explicit in §9.11. The new contributions of this paper are: the axiomatic characterization of perspective codes; the rigorous formulation and proof of the four structural theorems (rateless prefix, logarithmic UEP, amplitude covertness, holographic addressing); the gamma closed-form capacity (Theorem 9.2); the exponential precision law of the sign layer (Proposition 9.5); the multi-ledger accounting (§9.6); the dual-channel independent code (Theorem 9.7); the orthogonal isolation of the amplitude and sign layers (Theorem 9.9); and the exact independent verification of every numerical assertion.

§2 Axioms and Context

Notation. $H_r = \sum_{k=1}^{r} 1/k$; Δ^{m-1} is the m-dimensional probability simplex; e_k is the k-th standard basis vector; $Q(x) = \frac{1}{2}\operatorname{erfc}(x/\sqrt{2})$ is the Gaussian tail function; $\sigma_0 := a_{n+1} = 1/((n+1)H_{n+1})$ is the noise calibration unit (the weakest coordinate level); $\gamma_k := (a_k/\sigma)^2$ is the signal-to-noise ratio of coordinate k. Throughout, n is the number of message bits and $m = n + 1$ the number of coordinates.

Axiom P1 (orthogonal branching). The system state is a finite-dimensional nonnegative vector P; each update compresses the old mass globally by a factor $\lambda \in (0,1)$ and places the ceded mass $1-\lambda$ on a new coordinate orthogonal to all old ones: $P \mapsto \lambda P \oplus (1-\lambda)\, e_{\mathrm{new}}$.

Axiom P2 (normalization conservation). The total mass of the state is identically 1; the compression ratio at step r is $\lambda_r = H_r/H_{r+1}$, equivalently $1-\lambda_r = 1/((r+1)H_{r+1})$.

Axiom P3 (perfect memory). An update never changes the relative proportions among the old coordinates: for $k \le r$, $P_k^{(r+1)}/P_k^{(r)} = \lambda_r$, independent of k.

Axiom P4 (projective readout). Physical readout passes only through two projections: the sign $\operatorname{sgn}(\cdot)$ and the squared modulus $|\cdot|^2$. The message is encoded exclusively in the signs; the squared modulus is a public carrier.

Remark (the logical status of Axioms P2 and P3). P3 is in fact a direct corollary of P1: the update $P \mapsto \lambda P \oplus (1-\lambda)\, e_{\mathrm{new}}$ already compresses all old coordinates by the same factor ($P_k^{(r+1)}/P_k^{(r)} = \lambda$ for every old coordinate); it is listed separately only to emphasize the perfect-memory property. The “total mass identically 1” clause of P2 holds automatically under P1 and the initial condition $\|P^{(1)}\|_1 = 1$, since $\|\lambda P \oplus (1-\lambda)\, e\|_1 = \lambda\|P\|_1 + (1-\lambda) = 1$; the substantive content of P2 is the compression law $\lambda_r = H_r/H_{r+1}$ (equivalently $1-\lambda_r = 1/((r+1)H_{r+1})$).

Proposition 2.1 (unique solution of the axioms). The sequence of states satisfying P1–P3 with initial state $P^{(1)} = (1)$ is unique; it is the family of harmonic prefix vectors

$$P_k^{(r)} = \frac{1/k}{H_r}, \qquad 1 \le k \le r.$$

Proof. Induction on r. The case $r = 1$ is trivial. Suppose $P^{(r)}$ has been determined. By P1 and P3, the first r coordinates of $P^{(r+1)}$ are $\lambda_r P^{(r)}$, and the $(r+1)$-st coordinate is $1 - \lambda_r$; by P2, $\lambda_r = H_r / H_{r+1}$, whence

$$P_k^{(r+1)} = \frac{H_r}{H_{r+1}} \cdot \frac{1/k}{H_r} = \frac{1/k}{H_{r+1}} \quad (k \le r), \qquad P_{r+1}^{(r+1)} = 1 - \lambda_r = \frac{1/(r+1)}{H_{r+1}},$$

in agreement with the definition. ■

Lemma 2.1.1 (compatibility of sign injection). The modulus sequence $|E(b)_k| = a_k$ of a signed codeword still obeys the same recursion of P1–P3 (the signs attach coordinate by coordinate, and the evolution of the moduli coincides with the axiomatic states). Hence every modulus/proportion conclusion derived from the axioms holds for the moduli of signed codewords; the peeling recursions of Theorem 4.1(ii) and Theorem 5.1, acting on the two channels $|E|$ and sgn, invoke this lemma.

Definition 2.2 (the context $\mathscr{C}$). The public context shared by transmitter and receiver consists of: (i) the envelope table $a_k = (1/k)/H_{n+1}$ $(1 \le k \le n+1)$; (ii) the compression law $\lambda_r = H_r / H_{r+1}$; (iii) the set of prime positions among the coordinate indices (used for frame synchronization, Theorem 8.3). The context contains no message and may be broadcast publicly.

§3 Encoding

Definition 3.1 (perspective encoding). Fix a message length n. The encoding of a message $b = (b_1, \ldots, b_n) \in \{0,1\}^n$ is

$$E(b) \in \mathbb{R}^{n+1}, \qquad E(b)_1 = +a_1, \qquad E(b)_{k+1} = (-1)^{b_k} a_{k+1} \ (1 \le k \le n),$$

where $a_k = (1/k)/H_{n+1}$. Coordinate 1 is always positive and is called the anchor; coordinate $k+1$ carries the bit b_k ($0 \mapsto +1$, $1 \mapsto -1$; mapping table M.1).

Remark 3.1.1 (relation to the recursive generation). Inject the sign vector $s_k = \pm 1$ of Definition 3.1 into the recursion of Axioms P1–P3: $P^{(r+1)} = \lambda_r P^{(r)} \oplus (1-\lambda_r) s_{r+1} e_{r+1}$; expanding (the product telescopes) yields exactly $E(b)$. Thus the perspective code is not an afterthought applied to harmonic vectors but the direct continuation of the same generative grammar into signed algebra: the compression ratio λ_r determines the geometry (the angle spectrum of §7), while the signs s_k determine the message. The two are further separated into two independent channels in §9.7.

Worked example: the layer-by-layer states for $b = (1,0,1)$ (exact fractions):

Layer r	H_{r+1}	**State $E(b_{1..r})$ (coordinates 1..$r+1$)**
1	3/2	$\langle \frac{2}{3}, -\frac{1}{3} \rangle$
2	11/6	$\langle \frac{6}{11}, -\frac{3}{11}, \frac{2}{11} \rangle$
3	25/12	$\langle \frac{132}{275}, -\frac{66}{275}, \frac{44}{275}, -\frac{33}{275} \rangle$

Check: at layer $r = 3$, $H_4 = 25/12$, and the coordinates are $(1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4})/(25/12) = \langle \frac{12}{25}, -\frac{6}{25}, \frac{4}{25}, -\frac{3}{25} \rangle$, which over the common denominator 275 gives the table. The sum of squared moduli $\sum_k a_k^2 = H_4^{(2)}/H_4^2 = 41/125$ is independent of the message—the minimal instance of Theorem 6.3.

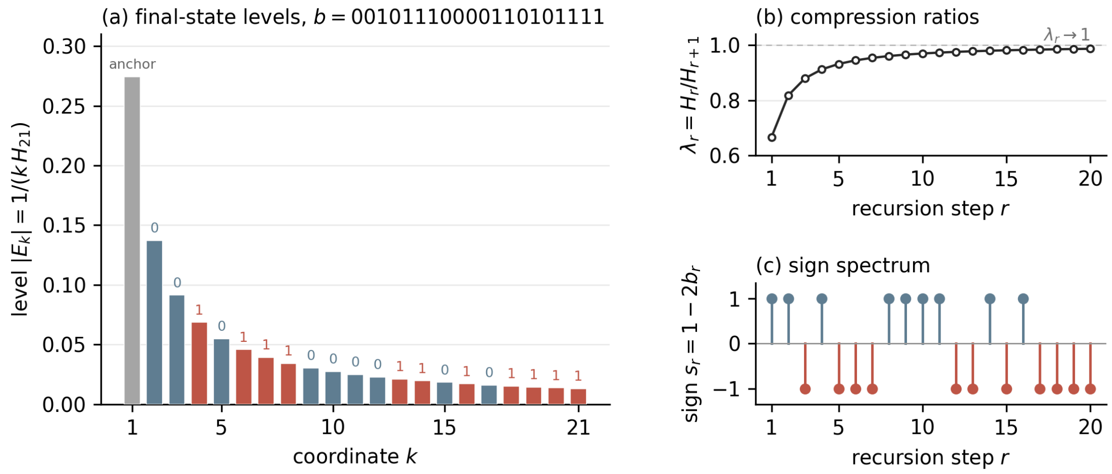

Figure 1

Figure 1. Structure of the perspective code for a 20-bit message (b = 00101110000110101111). Left: the levels $|E_k| = 1/(kH_{21})$ of the final state, red/blue distinguishing bits 1/0 by sign, grey for the anchor; note that the envelope is independent of the message (Theorem 6.3). Right: the freezing curve of the compression ratios λ_r (top, $\lambda_r \to 1$) and the sign spectrum (bottom).

§4 Decodability

Theorem 4.1 (unique decodability and dual-mode decoding). The map $E : \{0,1\}^n \to \mathbb{R}^{n+1}$ is injective, and each of the following two decoding modes recovers the message exactly:

i. Direct-read mode: $b_k = 1 \Leftrightarrow E(b)_{k+1} < 0$;

ii. Peeling mode: proceeding from the last coordinate backwards, read out the bit by $s_{r+1} = \operatorname{sgn}(P_{r+1}{}^{(r+1)})$, then recover the previous layer's state by the perfect-memory law $P_k{}^{(r)} = P_k{}^{(r+1)}/\lambda_r$ $(k \le r)$, and recurse down to $r = 1$.

Proof. Injectivity: if $b \neq b'$ the two sign vectors differ in some coordinate, and since $a_k > 0$, $E(b) \neq E(b')$. (i) is immediate from Definition 3.1. The correctness of (ii) rests on Axiom P3 (the recursive form of Proposition 2.1): $P_k{}^{(r+1)} = \lambda_r P_k{}^{(r)}$ $(k \le r)$. In the noiseless case λ_r is known from the context (Definition 2.2(ii)); equivalently, it can be read in one step from the modulus of the new coordinate: $\lambda_r = 1 - |P_{r+1}{}^{(r+1)}|$. For signed codewords the peeling recursion acts on the moduli $|E(b)_k|$ (Lemma 2.1.1), the signs being read out layer by layer by sgn. Thus $P^{(r)}$ is recovered exactly by dividing the first r coordinates of $P^{(r+1)}$ by λ_r; the sign readout at each layer gives $s_{r+1} = (-1)^{b_r}$. Induction down to $r = 1$. ■

Remark 4.1.1 (the division of labor between the two modes). The direct-read mode treats each coordinate independently, addressing one bit in $O(1)$; it has the strongest distortion immunity (§5). The peeling mode uses the envelope and the compression law for a global consistency check and is the carrier of the dual-channel decoding of §9.7. The two modes are equivalent in the absence of noise; under noise, the erasure decoding (Theorem 8.2, Algorithm A.4) is built on the direct-read mode, while the peeling mode (Mode II) is the exact-arithmetic carrier, whose noise/rounding sensitivity is quantified in Remark 9.7.1 (the error is amplified by $\prod_r \lambda_r^{-1}$).

Theorem 4.2 (holographic addressing). For any index set $S = \{i_1 < \cdots < i_t\} \subseteq \{1, \ldots, n\}$, the substring $b_S = (b_{i_1},\ldots,b_{i_t})$ can be decoded from the coordinate subset $\{i_j + 1\}$ by $O(t)$ sign readouts, with complexity independent of n; no other coordinate needs to be accessed.

Proof. Theorem 4.1(i) applies coordinate by coordinate: $b_{i_j} = \mathbb{1}[\,E_{i_j+1}<0\,]$, and each readout touches a single coordinate. ■

Remark 4.2.1. The word “holographic” is used here in a restricted sense: each coordinate carries one bit independently (rather than each fragment containing a miniature of the whole picture). What genuinely approaches the holographic meaning is the other direction of the perfect-memory law: the final state $E(b)$, inverted layer by layer, reproduces all prefix states (the peeling recursion of Theorem 4.1(ii)), i.e., the whole is written into every layer in compressed form. The two directions together give a two-way addressability: “any subset reads a substring; any final state replays the full history.”

§5 Geometric Immunity

Theorem 5.1 (immunity to global scaling). For every $c > 0$, both decoding modes applied to $c \cdot E(\mathbf{b})$ yield the same message as applied to $E(\mathbf{b})$.

Proof. Direct-read: $\operatorname{sgn}(cE_k) = \operatorname{sgn}(E_k)$ since $c > 0$. Peeling: scaling is self-consistent within the recursion—dividing $cP_k^{(r+1)}$ by the same λ_r gives $cP_k^{(r)}$, and the sign readout is unchanged; here λ_r is taken from the public context (Definition 2.2(ii)), or equivalently read off the magnitude of the new coordinate via $\lambda_r = 1 - |P_{r+1}^{(r+1)}|$ (the magnitude channel), and the scaling $c > 0$ is self-consistent for both readout channels. ■

Theorem 5.2 (immunity to per-coordinate positive distortion). For arbitrary strictly positive diagonal gains $d_1, \ldots, d_{n+1} > 0$, the direct-read mode applied to $(d_k E(\mathbf{b})_k)$ still recovers $\mathbf{b}$ exactly. That is, sign encoding is strictly immune to any unknown per-coordinate positive-gain channel.

Proof. The sign of $d_k E_k$ equals the sign of E_k because $d_k > 0$, and Theorem 4.1(i) reads only signs. ■

Remark 5.2.1. Theorem 5.2 is the strongest robustness statement for sign-layer coding: the channel may impose arbitrary unknown positive gains on every coordinate (attenuation, amplification, even extremely unbalanced fading across coordinates), and decoding emerges unscathed. The cost is stated by Theorem 6.2: coordinates with lower levels are more easily sign-flipped by noise—the robustness is against deterministic distortion, not random noise.

Theorem 5.3 (projective completeness). The encoding remains injective in the projective space $\mathbb{RP}^n$: $E(\mathbf{b})$ and $E(\mathbf{b}')$ ($\mathbf{b} \neq \mathbf{b}'$) do not belong to the same projective class; equivalently, the sign pattern of the normalized $E(\mathbf{b})/\|E(\mathbf{b})\|$ determines $\mathbf{b}$ uniquely.

Proof. Suppose $E(\mathbf{b}') = \mu E(\mathbf{b})$ with $\mu \neq 0$. Taking signs coordinatewise: $\operatorname{sgn}(\mu)\operatorname{sgn}(E(\mathbf{b})_k) = \operatorname{sgn}(E(\mathbf{b}')_k)$. At coordinate 1: $E(\mathbf{b})_1 = E(\mathbf{b}')_1 = a_1 > 0$, forcing $\mu > 0$; then all signs agree, so $\mathbf{b} = \mathbf{b}'$. ■

§6 Rateless Prefix, Logarithmic UEP, and Covertness

Theorem 6.1 (the rateless prefix property). Let $m \leq n$. The restriction of the full-length encoding $E(\mathbf{b}_{1..n}) \in \mathbb{R}^{n+1}$ to the first $m + 1$ coordinates differs from the prefix message's own encoding $E(\mathbf{b}_{1..m}) \in \mathbb{R}^{m+1}$ only by a global positive factor:

$$E(\mathbf{b}_{1..n})_{\{1,\ldots,m+1\}} = \frac{H_{m+1}}{H_{n+1}}\,E(\mathbf{b}_{1..m}).$$

Consequently: every prefix is a legal codeword; the prefix codeword encodes exactly the prefix message; and decoding may proceed at any prefix length without knowing n.

Proof. For $k \leq m + 1$,

$$E(\mathbf{b}_{1..n})_k = s_k\,\frac{1/k}{H_{n+1}} = \frac{H_{m+1}}{H_{n+1}} \cdot s_k\,\frac{1/k}{H_{m+1}} = \frac{H_{m+1}}{H_{n+1}}\,E(\mathbf{b}_{1..m})_k,$$

where $s_1 = +1$ and $s_k = (-1)^{b_{k-1}}$ $(k \geq 2)$ agree on both sides. The positive factor $H_{m+1}/H_{n+1} > 0$ does not affect decoding, by Theorem 5.1. ■

Remark 6.1.1 (comparison with fountain codes). LT codes [1] use a random degree distribution to achieve “arbitrarily many encoded packets, decodable once enough are collected”; the perspective code uses a deterministic prefix structure to achieve “any prefix length, decodable at any time.” The redundancy of the former lies between packets; that of the latter lies in the level ladder between coordinates (Theorem 6.2). The compression ratio $H_{m+1}/H_{n+1} < 1$ decreases as the prefix gets shorter—a shorter prefix has a weaker but structurally identical signal; this is the geometric price of ratelessness.

Theorem 6.2 (logarithmic unequal error protection). Over the AWGN channel $\mathbf{r} = E(\mathbf{b}) + \mathbf{N}$ with $\mathbf{N} \sim \mathcal{N}(0,\sigma^2 I_{n+1})$, the direct-read error probability of coordinate k is

$$\varepsilon_k = Q\left(\frac{a_k}{\sigma}\right) = Q\left(\frac{1}{k\,H_{n+1}\,\sigma}\right),$$

strictly increasing in k; the ratio of the SNRs of the strongest coordinate (the anchor $k = 1$, which carries no message) and the weakest coordinate ($k = n + 1$) is $(n+1)^2$, while within the message coordinates ($2 \leq k \leq n + 1$) the SNR dynamic range is $(n+1)^2/4$ (an amplitude dynamic range of $(n+1)/2$). Calibrated at $\sigma = \sigma_0$: $\varepsilon_k = Q((n+1)/k)$, degrading smoothly from $Q(150.5) < 10^{-100}$ at $k = 2$ to $Q(1) = 0.1587$ at $k = n + 1$. The protection strength is graded logarithmically in the coordinate:

$$\ln(1/\varepsilon_k) \approx \frac{(n+1)^2}{2k^2}.$$

Proof. A decoding error at coordinate k occurs if and only if N_k crosses the sign boundary, i.e., $|N_k| > a_k$ with the opposite direction, which has probability $Q(a_k/\sigma)$. Since $a_k = (1/k)/H_{n+1}$ is strictly decreasing in k and Q is monotone, ε_k is strictly increasing. The SNR is $\gamma_k = (a_k/\sigma)^2 \propto 1/k^2$, with head-to-tail ratio $((n+1)/1)^2 = (n+1)^2$ over all coordinates and $((n+1)/2)^2 = (n+1)^2/4$ over the message coordinates. The asymptotic $-\ln Q(x) \approx x^2/2$ gives the logarithmic grading formula. ■

Theorem 6.3 (amplitude covertness). The envelope vector $|E(\mathbf{b})| = (a_1,\ldots,a_{n+1})$ is independent of $\mathbf{b}$; equivalently, for any observer who sees only the squared modulus (or the modulus), $I(|E|;\mathbf{b}) = 0$, with exact equality (not asymptotically, not in a computational sense). The message information is carried entirely in the sign layer.

Proof. $|E(\mathbf{b})_k| = a_k$ is a constant vector independent of $\mathbf{b}$; hence the distribution of $|E|$ is independent of the distribution of $\mathbf{b}$, and the mutual information is zero. ■

Remark 6.3.1 (duality with Theorem 5.2). Theorem 5.2 establishes the strict immunity of the sign layer to positive distortion; Theorem 6.3 establishes the zero leakage of the amplitude layer to an eavesdropper. Together they delimit the security–robustness structure of the perspective code: both secrecy and fragility are concentrated in the single degree of freedom of the sign (phase). Contrast §9.8: once the amplitude is used to carry the message (PAM), both properties fail simultaneously.

§7 The Angle Spectrum

The square-root embedding $z_k = \sqrt{|E_k|}$ (together with the signs) sends the encoding to the unit sphere; the spherical angle θ_r between successive layer states satisfies $\cos\theta_r = \sqrt{\lambda_r}$ and $\tan\theta_r = 1/\sqrt{(r+1)H_r}$ (Theorem 6.2 of the companion paper; the same proof holds under the axioms of §2 here). Two clarifications are in order: (i) successive layer states have different dimensions ($z^{(r)} \in \mathbb{R}^r$, $z^{(r+1)} \in \mathbb{R}^{r+1}$), and the inner product is computed after embedding $z^{(r)}$ into $\mathbb{R}^{r+1}$ by appending a trailing zero; (ii) since $z_k^{(r+1)} = \sqrt{\lambda_r}\, z_k^{(r)}$ for $k \le r$ and $\|z^{(r)}\| = 1$, one has $\langle z^{(r)}, z^{(r+1)}\rangle = \sqrt{\lambda_r}\,\|z^{(r)}\|^2 = \sqrt{\lambda_r}$, hence $\sin\theta_r = \sqrt{1-\lambda_r}$ and therefore $\tan\theta_r = \sqrt{(1-\lambda_r)/\lambda_r} = 1/\sqrt{(r+1)H_r}$.[1] This section organizes this geometric observation into three statements in coding-theoretic language.

Proposition 7.1 (the completeness bijection of the angle spectrum). The following three kinds of data determine one another uniquely: (i) the message $\mathbf{b} \in \{0,1\}^n$; (ii) the sign sequence $(s_1,\ldots,s_{n+1})$ (with $s_1 = +1$); (iii) the direction of the normalized codeword $E(\mathbf{b})/\|E(\mathbf{b})\| \in S^n$. In particular, the message space is in bijection with the sign cells of the first orthant of the sphere, 2^n cells in total.

Proof. (i)$\Leftrightarrow$(ii) is Definition 3.1; (ii)$\Rightarrow$(iii): the direction is uniquely determined by the sign pattern together with the public envelope; (iii)$\Rightarrow$(ii): Theorem 5.3. ■

Proposition 7.2 (purely rational implementation, Fourier-free). The encoding and both decoding modes involve only addition, multiplication, and comparison of rational numbers; all intermediate quantities (states, envelopes, compression ratios) are rational, with numerator and denominator bit-width $O(n\log n)$ bits (the denominator $\operatorname{den}(H_{n+1}) \mid \operatorname{lcm}(1..n+1) = e^{\psi(n+1)}$). Decoding requires no trigonometric functions and no Fourier transform.

Proof. The a_k of Definition 3.1 are rational; direct-read is a comparison; peeling is rational multiplication and division. Bit-width: after bringing H_{n+1} to a common denominator, the denominator divides $\operatorname{lcm}(1..n+1)$, whose bit-length is $\psi(n+1)/\ln 2 = O(n)$; after bringing the n coordinates to a common denominator the total width is $O(n\log n)$ (each numerator and denominator has $O(n)$ bits, and $\psi(x) \sim x$ by the prime number theorem). ■

Remark 7.2.1. The angle-spectrum view (θ_r, Θ_r) corresponds physically to phase readout; it is mathematically complete (Proposition 7.1) and engineering-optional—the existence of the direct-read mode allows a receiver to dispense with angle estimation entirely. The geometric language of the angle spectrum is of the same family as information geometry [9], but none of its curvature machinery is needed. This forms a gradient with the q-phase generalization of §9.1: signs ($q = 2$) are the most robust, while phase ($q \to \infty$) has the highest capacity (Table B.7).

§8 Contextual Redundancy: Three Theorems

The context $\mathscr{C}$ (Definition 2.2) is not a passive parameter but a repository of redundancy. The three theorems of this section convert the envelope, envelope consistency, and the prime positions into error-control resources.

Theorem 8.1 (the all-pilot property). Every coordinate of the perspective code is its own pilot: the noiseless envelope $|E_k| = a_k$ is completely known to the receiver (Definition 2.2(i)), so channel-gain estimation requires no pilot overhead; within the coherence time, the per-coordinate gain estimate $\hat{d}_k = |r_k|/a_k$ is asymptotically unbiased, with an analytically known bias (a closed form is given in the proof).

Proof. Write $r = d \cdot a \cdot s + N$ with $d > 0$ the channel gain, $s = \pm 1$, and $N \sim \mathcal{N}(0,\sigma^2)$, and set $x := d\,a/\sigma > 0$. The absolute value folds the negative half-mass onto the positive half-axis, so $\hat{d}$ is strictly positively biased:

$$\mathrm{E}\,|r| - d\,a = 2\sigma\varphi(x) - 2d\,a\,Q(x) = \sqrt{\tfrac{2}{\pi}}\,\sigma\,x^{-2}e^{-x^2/2}(1 + O(x^{-2})) \qquad (x \to \infty),$$

where φ is the standard normal density (the asymptotic form follows from the Mills-ratio expansion $Q(x) = \varphi(x)(x^{-1} - x^{-3} + O(x^{-5}))$). Hence on strong coordinates ($x \gg 1$) the bias is exponentially small (rather than polynomially small), and $\hat{d}_k$ is asymptotically unbiased; on weak coordinates ($x = O(1)$; e.g., at $x = 1$ the bias is $\approx 0.166\,\sigma$) the bias is positive and non-negligible. Since a_k is known a priori, the bias term itself can be subtracted analytically, so the all-pilot property (zero pilot overhead) is unaffected. ■

Theorem 8.2 (envelope-consistency erasure). Define the per-coordinate envelope-consistency deviation

$$\kappa_k := \left| \frac{|r_k|}{a_k} - 1 \right|, \qquad 1 \le k \le n+1.$$

(Exact part) At $n = 300$, direct-read errors concentrate at coordinates with large κ: after declaring the ρ fraction of coordinates with the largest κ as erasures, the bit error rate on the retained coordinates, measured by 2×10^4 Monte Carlo trials (Table B.2; $\sigma = 0.5\sigma_0$, seed 20250825), drops from 3.29×10^{-3} (no erasure) to 3.07×10^{-4}

(erasing 5%), 1.17×10^{-4} (erasing 10%), and 4.16×10^{-5} (erasing 20%)—the first erasure (5%) gains about one order of magnitude, and thereafter each doubling of the erasure fraction lowers the error rate on the retained coordinates by about a factor of 3. (Criterion part) For coordinate k to be flipped yet remain unerased, the noise must simultaneously cross the boundary ($|N_k| > a_k$ in the opposite direction) and satisfy $|N_k| \approx 2a_k$ (to keep κ_k small): $|r_k|/a_k = |1 + N_k/(\pm a_k)| \approx 1$ with r_k of flipped sign forces $N_k \approx -2s_k a_k$, whose probability density is suppressed relative to the boundary-crossing probability by a factor on the order of

$$\exp\left(-\frac{(2a_k)^2 - (a_k)^2}{2\sigma^2}\right) = e^{-3\gamma_k/2};$$

this explains the observed order-of-magnitude gains.

Proof. The precise meaning of the criterion part: the event {coordinate k is wrong and $\kappa_k < \delta$} requires N_k to lie in an interval of width $2\delta a_k$ centered at $-2s_k a_k$, whose Gaussian probability is $\approx 2\delta a_k \cdot \varphi(2a_k/\sigma)/\sigma$; its ratio to the unconditional error probability $Q(a_k/\sigma) \approx \varphi(a_k/\sigma)\,\sigma/a_k$ is

$$\frac{2\delta a_k^2}{\sigma^2}\, e^{-3\gamma_k/2} = 2\delta\gamma_k\, e^{-3\gamma_k/2},$$

which is exponentially small for strong coordinates (large γ_k). This estimate requires $\gamma_k \gg 1$ (strong coordinates) and $\delta\sqrt{\gamma_k} \ll 1$ (near-constant density over the interval); for the weak-coordinate tail ($\gamma_k \approx 1$) the ratio is not exponentially small, consistent with the disappearance of the marginal gain of 5% erasure beyond $\sigma \gtrsim 0.8\sigma_0$ in Table B.2. Erasure therefore concentrates the errors onto the declared-erased coordinates, and the error rate on the retained coordinates falls accordingly. The Monte Carlo numbers are in Table B.2 (Algorithm A.4, 2×10^4 trials); the standard treatment of erasure as a decoding resource is [10]. ■

Theorem 8.3 (prime frame synchronization). The set of coordinate indices k that are prime is public context (Definition 2.2(iii)); by Theorem 7.1 of the companion paper (k is prime if and only if $\mathrm{den}(H_k)$ acquires the new prime factor k; for the classical denominator properties of harmonic numbers see [11]), the prime structure is recorded in the denominators of the harmonic numbers. The receiver can use the prime coordinates as known structural anchors for frame alignment: under an unknown frame offset τ, matched filtering against the prime-position coordinate set gives a normalized correlation peak at correct alignment of $\sum_{p \le n+1} a_p^2 / \sum_k a_k^2$, which for large n tends to

$$\frac{\sum_p 1/p^2}{\pi^2/6} = P(2)/\zeta(2) = 0.452247\ldots \times 6/\pi^2 = 0.27493\ldots \approx 27.5\%.$$

Proof. The numerator of the correlation peak: the energy share of the prime coordinates is $\sum_{p \le n+1} a_p^2 / \sum_{k \le n+1} a_k^2 = (\sum_{p \le n+1} 1/p^2)/H_{n+1}^{(2)} \to (\sum_p p^{-2})/\zeta(2)$; here $\sum_p p^{-2} = 0.452247\ldots$ (the prime zeta function at 2) and $\zeta(2) = \pi^2/6$. The normalized correlation at offset τ is

$$R(\tau) = \frac{\sum_{p:\, p+\tau \le n+1} a_{p+\tau}^2}{\sum_k a_k^2},$$

which is monotonically decreasing in $|\tau|$; in the limit $n \to \infty$, $R(0) = P(2)/\zeta(2) = 0.27493\ldots$, $R(1) = 0.14862\ldots$ (54.1% of the aligned peak), $R(2) = 0.09729\ldots$ (35.4%), $R(10) = 0.02088\ldots$ (7.6%), and $R(\tau) \approx 6/(\pi^2\tau\ln\tau) \to 0$ for large τ. Frame synchronization rests on the contrast $R(0) - R(\tau)$, not on a vanishing misalignment expectation; the scarcity of the prime set (density $1/\ln n$) suppresses false peaks at large τ. ■

§9 Capacity Analysis

This section settles, level by level, the information-carrying capability of the perspective code over the AWGN channel $r = E(b) + N$, $N \sim \mathcal{N}(0,\sigma^2 I_{n+1})$. The noise is always calibrated in units of $\sigma_0 = a_{n+1} = 1/((n+1)H_{n+1})$, at which the coordinate SNRs are

$$\gamma_k = \left(\frac{a_k}{\sigma_0}\right)^2 = \left(\frac{n+1}{k}\right)^2, \qquad 1 \le k \le n+1,$$

a deterministic square ladder from $(n+1)^2$ down to 1.

9.1 The *q*-phase sign generalization

Proposition 9.1 (*q*-phase constellations). Generalize the sign degree of freedom of each coordinate to q equally spaced phases in the complex plane (mapping table M.2). The complex-channel convention is adopted as follows

$$y = \sqrt{\gamma}\, e^{i\varphi} + z, \qquad z \sim \mathrm{CN}(0,1),$$

where $\gamma = E_s/N_0$ is the symbol SNR and $N_0 = 1$ is the total variance of the complex noise (variance 1/2 per real dimension). Each coordinate carries $\log_2 q$ bits with decision margin $\pm\pi/q$; at coordinate SNR γ, the achievable mutual information of q-phase PSK saturates at $\log_2 q$ as γ grows. Under this convention, the $q = 2$ row of Table B.7 is the complex-channel equivalent $C_{\mathrm{BPSK}}(2\gamma)$ of real-axis BPSK (the signal is confined to the real axis, the noise split equally between the two dimensions), and the $q = 4$ row is $2\,C_{\mathrm{BPSK}}(\gamma)$ (two orthogonal BPSK components carrying one bit each), consistent with the note in M.2. Numerical integration (Table B.7): $q = 4$ reaches 2.00 bits (saturation) at $\gamma = 25$, $q = 8$ reaches 3.00 bits at $\gamma = 49$, and $q = 16$ reaches 3.98 bits at $\gamma = 100$; at $n = 1000$, the total q-phase rates over all coordinates (complex channel, $\gamma_k = ((n+1)/k)^2$) for $q = 2, 4, 8, 16$ are 938.9/1677.2/2084.6/2285.8 bits respectively, where $q = 4$ is exactly twice the sign-layer total $C_{\mathrm{BPSK}} = \sum_{k=1}^{n+1} C_{\mathrm{BPSK}}(\gamma_k) = 838.6$ bits (the sum includes the anchor coordinate $k = 1$, whose contribution ≈ 1 bit is asymptotically negligible); against the complex-coordinate total capacity $2C_{\mathrm{Sh}} = 3257.3$ bits, $q = 16$ already covers 70.2%.

9.2 A gamma closed form for the total capacity

Theorem 9.2 (capacity closed form). The sum of the capacities of the coordinates used independently with independent Gaussian inputs is

$$C_{\mathrm{Sh}}(n) = \frac{1}{2}\sum_{k=1}^{n+1} \log_2(1+\gamma_k) = \frac{1}{2}\log_2 \prod_{k=1}^{m}\left(1+\frac{m^2}{k^2}\right), \qquad m := n+1,$$

with the gamma-function closed form

$$C_{\mathrm{Sh}}(n) = \frac{1}{2}\log_2 \frac{|\Gamma(m+1+im)|^2}{|\Gamma(1+im)|^2\,(m!)^2}$$

and the asymptotic expansion

$$C_{\mathrm{Sh}}(n) = \frac{\pi + 2\ln 2}{4\ln 2}\,m - \frac{1}{2}\log_2 m + \left(\frac{1}{4} - \frac{1}{2}\log_2(2\pi)\right) + O(1/m) = 1.6330900355\ldots \cdot m - \frac{1}{2}\log_2 m - 1.0757\ldots + O$$

Proof. The first equality is the sum of the per-coordinate Gaussian capacities $\frac{1}{2}\log_2(1+\gamma)$ ([3][5]). Closed form:

$$\prod_{k=1}^{m}\left(1+\frac{m^2}{k^2}\right) = \prod_{k=1}^{m}\frac{(k+im)(k-im)}{k^2} = \frac{|\Gamma(m+1+im)|^2/|\Gamma(1+im)|^2}{(m!)^2},$$

using the product characterization $\prod_{k=1}^{m}(k+z) = \Gamma(m+1+z)/\Gamma(1+z)$ of Γ and $|w|^2 = w\bar{w}$. Asymptotics: apply the Stirling expansion to the closed form. From $|\Gamma(1+im)|^2 = \pi m/\sinh(\pi m)$ and $\ln\Gamma(z) = (z-\frac{1}{2})\ln z - z + \frac{1}{2}\ln(2\pi) + O(1/z)$ with $z = m+1 \pm im$,

$$\sum_{k=1}^{m}\ln\left(1+\frac{m^2}{k^2}\right) = \left(\frac{\pi}{2} + \ln 2\right)m - \ln(2\pi m) + \frac{\ln 2}{2} + O(1/m),$$

and dividing both sides by $2\ln 2$ yields the stated expansion. The leading coefficient admits an interpretive reading: an integral approximation to $\sum_k \ln(1+m^2/k^2)$ gives $m\int_0^1 \ln(1+t^{-2})\,dt$; precisely,

$$\int_0^1 \ln(1+1/t^2)\,dt = \left[\,t\ln(1+1/t^2) + 2\arctan t\,\right]_0^1 = \ln 2 + \frac{\pi}{2},$$

hence the leading term $m(\ln 2 + \pi/2)/(2\ln 2) = \frac{\pi + 2\ln 2}{4\ln 2}\,m$. Numerical verification (Table B.3): at $n = 100, 1000, 10000$, $C_{\mathrm{Sh}} = 160.5366,\ 1628.6637,\ 16324.8138$ bits, the sum and the closed form agreeing digit by digit; the expansion evaluated at $m = 1001$ gives 1628.6637, likewise digit by digit. ■

Remark 9.2.1 (the meaning of "1.63 bits per coordinate"). The per-coordinate average capacity exceeds 1 bit because the strong coordinates have very large γ_k: the capacity is concentrated at the head $k \ll m$. The corresponding account is settled in §9.4; its correct reading—the pricing of ladder depth, not an efficiency defect—is established by Proposition 9.5 and Remark 9.5.2. Two conventions for the efficiency η must be distinguished: (i) the noiseless-limit convention

$$\eta_0 := \frac{1}{1.6331} = \frac{4\ln 2}{\pi + 2\ln 2} = 0.612336,$$

the ratio of the sign layer's asymptotic at most 1 bit per coordinate to the Shannon limit 1.6331; (ii) the operating-point convention at $\sigma = \sigma_0$,

$$\eta(\sigma_0) := \frac{0.8380}{1.6331} = 0.513,$$

the ratio of the sign-constrained capacity C_{BPSK} to the Shannon capacity (the convention of Table B.3). Throughout this paper, η takes convention (i).

9.3 The reliable-information ladder

Four usable rates, degrading step by step from soft information to hard decision (n = 100/1000/10000, Table B.3; all verified by independent numerical integration):

- C_{Sh} (Gaussian codebook, per-coordinate soft information): 160.54/1628.66/16324.81 bits, asymptotically $1.6331n$;
- C_{BPSK} (sign-constrained capacity, $\sum_{k=1}^{n+1} C_{\mathrm{BPSK}}(\gamma_k)$, the sum including the anchor coordinate $k = 1$, whose contribution ≈ 1 bit is asymptotically negligible; $C_{\mathrm{BPSK}}(\gamma) = 1 - \mathrm{E}_Z \log_2(1 + e^{-2\gamma-2\sqrt{\gamma}\,Z})$, $Z \sim \mathcal{N}(0,1)$): 84.4/838.6/8380.7 bits, asymptotically $\approx 0.838n$;
- R_{hard} (sum of hard-decision BSC capacities, $\sum_k [1 - H_2(Q(\sqrt{\gamma_k}))]$): 78.0/775.8/7753.4 bits, asymptotically $\approx 0.775n$;
- $R_{\mathrm{eras\,20\%}}$ (hard-decision capacity sum over the $\lceil 0.8m \rceil$ highest-SNR coordinates retained): 69.4/688.6/6881.1 bits, asymptotically $\approx 0.688n$.

How to read the ladder: $C_{\mathrm{Sh}}/C_{\mathrm{BPSK}} \approx 1.95$ is the price of the sign constraint; $C_{\mathrm{BPSK}}/R_{\mathrm{hard}} \approx 1.081$ is the price of hard decision; $R_{\mathrm{hard}}/R_{\mathrm{eras\,20\%}} \approx 1.126$ is the price of 20% of the coordinates being unavailable (while structural redundancy is still required). The three ratios are computed from the asymptotic coefficients and vary with n by no more than a few percent—the ladder is scale-free.

9.4 Energy scaling

Proposition 9.4 (the energy ledger). The total codeword energy is $E(n) = \sum_k a_k^2 = H_{n+1}^{(2)}/H_{n+1}^2 \to (\pi^2/6)/\ln^2(n{+}1)$; under the hard-decision reliability constraint (the weakest-coordinate error rate ε_{n+1} held fixed, i.e., $\sigma \propto \sigma_0$), the energy-per-bit to noise ratio is

$$\frac{E_b}{N_0} = \frac{c^2\,\pi^2\,(n+1)^2}{12\,n}\,(1 + o(1)), \qquad c := Q^{-1}(\varepsilon_0),$$

where ε_0 is the target error rate of the weakest coordinate. Taking $\varepsilon_0 = 10^{-2}$ ($c \approx 2.3263$, $c^2 \approx 5.4119$): E_b/N_0 = 26.55/36.49/46.49 dB at n = 100/1000/10000 (exact values, Table B.5)—linear growth in n, i.e., $+10$ dB per decade. The anchor coordinate's energy share is $a_1^2/E(n) \to \dfrac{1}{\pi^2/6} = \dfrac{6}{\pi^2} \approx 60.8\%$. The accounting of this proposition is the "sign layer alone" (empty envelope) one; the correct reading of the ledger is given in Remark 9.5.2.

Proof. $\sum_k a_k^2 = H_{n+1}^{(2)}/H_{n+1}^2$, with $H^{(2)} \to \pi^2/6$ and $H \sim \ln$. Anchor share: $a_1^2/E = (1/H_{n+1}^2)/(H_{n+1}^{(2)}/H_{n+1}^2) = 1/H_{n+1}^{(2)} \to 6/\pi^2$. For E_b/N_0: reliability requires the weakest coordinate's error rate $Q(a_{n+1}/\sigma)$ not to exceed the target ε_0, i.e., $a_{n+1}/\sigma = c = Q^{-1}(\varepsilon_0)$, equivalently $\sigma = \sigma_0/c$; with $N_0 = 2\sigma^2$ and $E_b = E/n$, substituting $E \approx (\pi^2/6)/H_{n+1}^2$ and $\sigma_0^2 = 1/((n{+}1)^2 H_{n+1}^2)$ yields

$$\frac{E_b}{N_0} = \frac{E/n}{2\sigma_0^2/c^2} = \frac{\pi^2(n+1)^2c^2}{12\,n},$$

where the constant c is completely determined by the target error rate; $\varepsilon_0 = 10^{-2}$ corresponds to $c = Q^{-1}(10^{-2}) \approx 2.3263$, $c^2 \approx 5.4119$, and substitution gives the figures in the text (Table B.5). ■

Proposition 9.5 (the exponential precision law of the sign layer). Let coordinate k carry a bipolar sign, with per-coordinate SNR $\gamma_k = {a_k}^2/\sigma^2$ and hard-decision error rate $P_e(k) = Q(\sqrt{\gamma_k})$. Define the reliable decimal digits of the coordinate as $D_k := -\log_{10}P_e(k)$. Then:

i. (two-sided bounds) for every $\gamma_k \geq 0$,

$$D_k \geq \frac{\gamma_k}{2\ln 10} + \log_{10} 2;$$

and for $\gamma_k > 1$,

$$D_k \leq \frac{\gamma_k}{2\ln 10} + \frac{1}{2}\log_{10}(2\pi\gamma_k) - \log_{10}(1-\gamma_k^{-1}).$$

ii. (asymptotic slope) as $\gamma_k \to \infty$, $D_k = \dfrac{\gamma_k}{2\ln 10} + O(\log\gamma_k)$: the reliable digits grow linearly in the SNR with the universal slope $1/(2\ln 10) = 0.2171472410$; equivalently, the reliable precision $\varepsilon_k = 10^{-D_k}$ improves at least exponentially as the SNR increases linearly:

$$\varepsilon_k \leq \tfrac{1}{2}\,e^{-\gamma_k/2}.$$

iii. (constant incremental pricing) the SNR increment needed to raise D_k by one more digit tends to a constant: $\Delta\gamma_k = 2\ln 10 + o(1) = 4.6051701860 + o(1)$ $(\gamma_k \to \infty)$, independent of the coordinate k and of the digits already achieved; in the dB scale, the cost of each additional digit decreases with the operating point.

On the ladder $\gamma_k = \gamma_1/k^2$, so $D_k = \dfrac{\gamma_1}{2k^2\ln 10} + O(\log\gamma_k)$: the last column $-\ln\varepsilon_k \approx (n+1)^2/(2k^2)$ of Table B.1 is the leading term of this law at $\sigma = \sigma_0$. The same exponential law governs the grid precision of the dual-channel λ readout (Theorem 9.7): nearest-point readout on a grid of pitch δ has error rate exactly $2Q(\delta/(2\sigma))$.

Proof. $P_e(k) = Q(\sqrt{\gamma_k})$ is the standard hard-decision fact of the real bipolar channel (Table B.1 tabulates it coordinatewise at $\sigma = \sigma_0$). The standard Q-function bounds are used: for $x > 0$,

$$\frac{e^{-x^2/2}}{\sqrt{2\pi}}\left(\frac{1}{x} - \frac{1}{x^3}\right) \leq Q(x) \leq \frac{1}{2}e^{-x^2/2}.$$

Lower bound: $g(x) := Q(x) - \varphi(x)(x^{-1} - x^{-3})$ satisfies $g(+\infty) = 0$ and $g'(x) = -3\varphi(x)/x^4 < 0$, hence $g > 0$. Upper bound: $h(x) := \frac{1}{2}e^{-x^2/2} - Q(x)$ satisfies $h(0) = h(+\infty) = 0$, and $h'(x) = \varphi(x)(1 - \sqrt{\pi/2}\,x)$ has exactly one zero on $(0,\infty)$, positive before it and negative after, hence $h \geq 0$. Substituting $x = \sqrt{\gamma_k}$ and applying $-\log_{10}$ gives (i) (the upper bound requires $\gamma_k > 1$ so that $1 - \gamma_k^{-1} > 0$). Both ends of (i) share the leading term $\gamma_k/(2\ln 10)$ with

$O(\log\gamma_k)$ remainders, giving (ii); $\varepsilon_k \le \frac{1}{2}e^{-\gamma_k/2}$ is a direct rewriting of the upper bound. Solving $D_k(\gamma_k{+}\Delta) - D_k(\gamma_k) = 1$: by (ii), $\Delta/(2\ln 10) + O(\log(1{+}\Delta/\gamma_k)) = 1$, so $\Delta \to 2\ln 10$ as $\gamma_k \to \infty$, giving (iii). Grid remark: when the true value lies on a grid point, nearest-point decoding errs if and only if the coordinate noise crosses the decision boundary at distance $\delta/2$, an event of probability exactly $2Q(\delta/(2\sigma))$, the same law as (ii). ■

Remark 9.5.1 (flatness of reliable digits per joule). The lower bound of Proposition 9.5(i) gives, coordinate by coordinate, $D_k \ge \dfrac{e_k}{2\sigma^2 \ln 10} + \log_{10} 2$ (with $e_k = a_k^2$ the coordinate energy): the reliable decimal digits are proportional to the coordinate energy with the universal constant $\dfrac{1}{2\sigma^2 \ln 10}$, independent of k and n. Summed over the codeword:

$$\sum_{k=1}^{n+1} D_k \ge \frac{E}{2\sigma^2 \ln 10} + (n+1)\log_{10} 2,$$

and the bound is asymptotically tight: along $\sigma = \sigma_0$ as $n \to \infty$, the ratio of the exact sum to the energy-proportional leading term is 1.0251/1.0025/1.0003 (n = 100/1000/10000; exact sums $3712.7/3.5860 \times 10^5/3.5733 \times 10^7$ digits versus leading terms $3621.9/3.5769 \times 10^5/3.5724 \times 10^7$ digits, computed with log-domain Q-function evaluations). The implication: the distribution of energy over the ladder is exactly the distribution of delivered precision—the anchor holds $6/\pi^2$ of the energy and also delivers approximately $6/\pi^2$ of the reliable digits; no rung is wasted in the reliable-digits-per-joule currency. Pushing the error floor of the whole ladder down by one decimal order costs only a uniform additive $2\ln 10 \approx 4.6052$ (linear units) of SNR per coordinate.

Remark 9.5.2 (the correct reading of the energy ledger). The account of Proposition 9.4 prices a specific design choice: extending protection at the same target error rate ε_0 along the UEP ladder all the way down to the weakest coordinate $k = n + 1$. It is not an efficiency defect of the sign layer, for three reasons. (i) What this expenditure yields is a precision reserve: by Proposition 9.5, every linear increment of the account (adding $2\ln 10$ to γ) adds one reliable digit on every rung of the ladder—linear input, exponential output. (ii) By Remark 9.5.1, energy and delivered precision are strictly proportional across all rungs; no rung inside the ladder is wasted. (iii) By Theorem 9.9(iv), the same energy budget can simultaneously carry an independent amplitude-layer stream; charging all of the energy to the sign-layer bits is the accounting of the single-use case with an empty envelope, and the composite efficiency is given by Table B.8 ($\approx 95.0\%\, C_{\mathrm{Sh}}$). The genuine niche of the single-use case is precision- and security-oriented payloads (storage, holographic media, physical-layer security), where the correct unit of account is reliable digits per joule (Remark 9.5.1), not joules per bit.

9.5 Continuous capacity of the phase channel

Treat the dual of the sign degree of freedom—the continuous phase $\varphi \in [0, 2\pi)$—as a channel (received phase $\theta + \nu$, ν the phase noise). The capacity of the circular phase channel at SNR γ is asymptotically

$$C_{\mathrm{ph}}(\gamma) = \frac{1}{2}\log_2 \frac{4\pi\gamma}{e} + o(1) \qquad (\gamma \to \infty),$$

derived as follows: under complex AWGN the equivalent variance of the phase noise is $\sigma_{\mathrm{eff}}^2 = 1/(2\gamma)$ (the standard reduction of the perpendicular noise component by the amplitude); at high SNR the wrap-around effect is negligible, and the capacity is the difference between the differential entropy $\log_2(2\pi)$ of the uniform prior and the noise differential entropy $\frac{1}{2}\log_2(2\pi e\sigma_{\text{eff}}^2) = \frac{1}{2}\log_2(\pi e/\gamma)$, namely $\frac{1}{2}\log_2(4\pi\gamma/e)$. This curve connects the q-phase ladder of Proposition 9.1 into a continuous family (the saturation values of Table B.7 are its integer samples).

9.6 Multi-ledger accounting on fading channels

Introduce per-coordinate real Gaussian fading $G_k \sim \mathcal{N}(0,1)$ (i.i.d., independent of the message), $r_k = G_k E(b)_k + N_k$. This is the standard real-projection model of Rayleigh fading on the in-phase component: the real part of a complex gain $\mathcal{CN}(0,1)$ is exactly $\mathcal{N}(0,1)$; the random sign of G_k embodies the receiver's missing coherent phase reference. The chain rule gives the two-way expansion

$$I(r;b,G) = \underbrace{I(r;b)}_{\text{blind signs}} + \underbrace{I(r;G \mid b)}_{\text{learning fading from known message}} = \underbrace{I(r;G)}_{\text{total fading information}} + \underbrace{I(r;b \mid G)}_{\text{message with known fading}}.$$

Proposition 9.6 (exact values of the four ledgers). (i) $I(r;b) = 0$ exactly: given b, $G_k s_k \sim \mathcal{N}(0,1)$ is independent of s_k, so $r_k \mid b \sim \mathcal{N}(0,a_k^2+\sigma^2)$, and the marginal distribution carries no message information; (ii) $I(r;G|b) = C_{\text{Sh}}(n)$ exactly: given the signs, $G_k s_k$ is still a standard Gaussian variable, so coordinate k is a standard Gaussian channel with input $\mathcal{N}(0,a_k^2)$ and noise σ^2, of capacity $\frac{1}{2}\log_2(1+\gamma_k)$; summing over coordinates gives Theorem 9.2; (iii) $I(r;b|G) = \sum_k E_G\, C_{\text{BPSK}}(\gamma_k G^2)$ (given G, reception is coherent), numerically 62.7/624.5 bits ($n = 100/1000$, asymptotically $\approx 0.624n$; the sum includes the anchor coordinate $k = 1$, whose contribution is ≈ 0.99 bits and asymptotically negligible); (iv) $I(r;G) = C_{\text{Sh}}(n) - I(r;b|G)$ holds term by term exactly, numerically 97.8/1004.2 bits ($n = 100/1000$, asymptotically $\approx 1.009n$: the finite-n per-bit ratios are 0.978/1.004, the $O(\log n/n)$ correction falling almost entirely on this term); the identity closes: $0 + C_{\text{Sh}} = I(r;G) + I(r;b|G)$ term by term for every coordinate ($n = 1000$: $624.45 + 1004.21 = 1628.66 = C_{\text{Sh}}$, see Table B.6), and the asymptotic coefficients close as well: $0.624 + 1.009 = 1.633$, the linear coefficient of C_{Sh} (Theorem 9.2).

Proof. The arguments for (i)(ii)(iii), given above, are all standard Gaussian-channel identities. The core is the termwise identity (iv). Fix a coordinate and write the dimensionless quantity $u = r_k/a_k = Gs + Z/\sqrt{\gamma}$ (per-coordinate noise variance $\sigma^2 = 1/\gamma$, s uniform). On the one hand $u \sim \mathcal{N}(0,1+\sigma^2)$ (since Gs is standard Gaussian), so $h(u) = \frac{1}{2}\log_2(2\pi e(1+\sigma^2))$; on the other hand $u \mid G = g$ is the symmetric Gaussian mixture $\frac{1}{2}\mathrm{N}(g,\sigma^2) + \frac{1}{2}\mathrm{N}(-g,\sigma^2)$. Decompose its entropy via $p(u \mid g) = \varphi(u-g;\sigma^2)\cdot\frac{1}{2}(1+e^{-2gu/\sigma^2})$: the first factor contributes $\frac{1}{2}\log_2(2\pi\sigma^2) + \dfrac{(\sigma^2+2g^2)\log_2 e}{2\sigma^2}$; in the second factor, expanding separately over the two halves of the mixture and using the symmetry of Z,

$$E_p \log_2 \tfrac{1}{2}(1+e^{-2gu/\sigma^2}) = E_Z \log_2 \tfrac{1}{2}(1+e^{-2g^2\gamma - 2g\sqrt{\gamma}\,Z}) + g^2\gamma\log_2 e = -C_{\text{BPSK}}(g^2\gamma) + g^2\gamma\log_2 e,$$

and the $g^2\gamma\log_2 e$ terms of the two parts cancel exactly, giving the compact closed form

$$h(u \mid G = g) = \tfrac{1}{2}\log_2(2\pi e\sigma^2) + C_{\text{BPSK}}(g^2\gamma).$$

(Checks: $g = 0$ degenerates to the entropy of a single Gaussian; as $|g| \to \infty$ the mixture entropy equals the single-component entropy $+1$ bit, and $C_{\mathrm{BPSK}} \to 1$.) Therefore

$$I(u; G) = h(u) - E_G\, h(u \mid G) = \tfrac{1}{2} \log_2(1 + \gamma) - E_G\, C_{\mathrm{BPSK}}(\gamma G^2),$$

equal term by term to the difference of (ii) and (iii). Numerically, by Gauss–Hermite quadrature (with the C_{BPSK} lookup table cross-validated by adaptive quadrature): at $n = 100/1000$, (iii) $= 62.71/624.45$ and (iv) $= 97.82/1004.21$, summing to $160.54/1628.66 = C_{\mathrm{Sh}}$, closing to within quadrature precision. ■

9.7 Dual-channel independent coding

Theorem 9.7 (the compression-ratio channel). Free the compression ratio in the generation recursion from a constant into a message parameter: at layer r, λ_r is drawn from the 8-level set $\{j/9 : j = 1, \ldots, 8\}$ (mapping table M.3), carrying 3 bits per layer, independently of the sign channel (1 bit per layer). Decoding: in peeling mode, the modulus of the new coordinate at layer r is exactly $1 - \lambda_r$, so $\hat{\lambda_r} = 1 - |P_{r+1}{}^{(r+1)}|$ is read out in one step without iteration. Exact rational-arithmetic simulation: 501 coordinates (500 layers) carry 2000 bits over the two channels ($500 \times (1{+}3)$), with λ and the signs bit-exact throughout (Table B.9).

Proof. Independence: in the recursion $P^{(r+1)} = \lambda_r P^{(r)} \oplus (1{-}\lambda_r) s_{r+1} e_{r+1}$, the sign s_{r+1} and the compression ratio λ_r control the "direction" and the "proportion" respectively; given λ_r the sign readout is unaffected, and given the sign, λ_r is read from the modulus of the new coordinate in one step; the peeling recursion reads λ_r (modulus) first and the sign second, and the two do not interfere. Exactness under exact arithmetic: the 8 levels have minimum spacing 1/9 and decoding margin 1/18, and rational arithmetic has no rounding—in the 500-layer simulation both λ and the signs were bit-exact throughout (the largest coordinate denominator has 408 decimal digits, consistent with the bit-width of $\prod_r \mathrm{den}(\lambda_r) = \prod_r 9/\gcd(j_r, 9)$). ■

Remark 9.7.1 (numerical stability—the intrinsic cost of the dual channel). The peeling recursion amplifies the rounding (or noise) error of layer j by $\prod_{r \ge j} \lambda_r{}^{-1}$: when the λ_r are uniform over $\{j/9\}$, the expected amplification is $\exp(E \ln(1/\lambda)) = e^{0.8716} \approx 2.39$ per layer. Consequently float64 (52-bit mantissa) supports on average only about 41 layers (16 layers in the worst case $\lambda \equiv 1/9$), and 500 layers require a mantissa of about 629 bits. The dual-channel mode must therefore operate under the purely rational implementation of Proposition 7.2; on a physical AWGN channel, the reliability of the inner λ bits deteriorates by the same factor (open problem 3). The sign channel is unaffected: direct-read only compares signs (Theorem 4.1).

9.8 A Shannon-approaching modification and its price

Proposition 9.8 (the 2Q-PAM modification). Abandon the constraint that "the message lives only in the signs" and load each coordinate, according to its SNR, with a uniform $2Q_k$-PAM constellation: take the squared half-spacing $d^2 = \pi e/8$ (so that under the dense-constellation approximation $\log_2(2Q) - \mathrm{MI} \to \frac{1}{2} \log_2 (2\pi e \sigma^2/(4d^2)) = 1$ bit holds exactly), substitute into the average-energy constraint $d^2(4Q^2{-}1)/3 = \gamma\sigma^2$ to obtain $Q \approx \sqrt{6\gamma/(\pi e)}$, and round to integers: $Q_k = \lceil \sqrt{6\gamma_k/(\pi e)} \rceil$. Then the total rate is

$$R_{\mathrm{PAM}}(n) = \sum_{k=1}^{n+1} \log_2(2Q_k) - \text{shaping loss}, \qquad \text{shaping loss} = \tfrac{1}{2}\log_2(\pi e/6) = 0.2546 \text{ bits/coordinate (upper bound)},$$

and the measured rate reaches 94.9–95.0% of C_{Sh} (n = 100/1000/10000: 152.4/1547.0/15507.7 bits; the n = 100 row is 94.9%), with a residual gap of 0.080/0.082/0.082 bits per layer (Table B.8). Note that the bit-depth sum $\sum_k \log_2(2Q_k)$ = 250.4/2517.1/25204.1 bits given by the loading rule exceeds C_{Sh} (overloading): the actual rate is automatically truncated inside capacity by the per-coordinate mutual information $\mathrm{MI}(2Q_k,\gamma_k)$, with $\log_2(2Q_k) - \mathrm{MI} \to 1.0$ bit for large Q (the exact identity $\tfrac{1}{2}\log_2(24/(\pi e)) + \tfrac{1}{2}\log_2(\pi e/6) = 0.7454 + 0.2546 = 1$; measured 0.9971/1.0009 at $\gamma = 10^4/10^6$). The price: once the amplitude layer starts carrying information, Theorem 6.3 (covertness) and Theorem 5.2 (distortion immunity) fail simultaneously—a consequence of the amplitude channel being activated, not of any failure of the isolation. This modification is a direct instance of Theorem 9.9: the amplitude layer is loaded according to the capacity rule (the degree of freedom of conventional coding), the sign layer remains the perspective code, and the two streams are mutually independent.

Remark 9.8.1 (the origin of the shaping loss). The loss of uniform 2Q-PAM relative to a Gaussian codebook is the differential-entropy gap $\tfrac{1}{2}\log_2\dfrac{\pi e}{6} \approx 0.2546$ bits between the uniform and Gaussian distributions (the one-dimensional case of Forney–Wei shaping theory [7]; for the standard framework of PAM constellations and coded modulation see [12]); an outer shaping code recovers most of it. The residual gap consists of three parts: the per-coordinate shaping loss of the high-SNR coordinates (0.2546 bits each for $2Q_k > 128$), the BPSK–Gaussian gap of the low-SNR coordinates, and the effect of integer rounding; none of the three grows with n. The measured 94.9–95.0% efficiency shows that the SNR ladder $\gamma_k = ((n+1)/k)^2$ of the perspective code matches the capacity loading rule naturally, without water-filling iterations.

9.9 Orthogonal isolation of the amplitude and sign layers

Theorem 6.3 states that the envelope readout carries no message ($I(|E|;b) = 0$)—in the base code the envelope is an unoccupied orthogonal resource. This section develops the positive side: the unoccupied layer can be populated by an arbitrary independent message stream without disturbing the sign layer. Model: the real AWGN channel $y = x + z$, $z \sim \mathcal{N}(0,\sigma^2)$; the transmitted symbol decomposes as $x = s\,a$, where the amplitude $a \in \mathcal{A} \subset \mathbb{R}_{>0}$ carries a message stream w_a and the sign $s \in \{\pm 1\}$ is uniform and carries a message stream w_s; the two streams are statistically independent and arbitrary in content—they need not carry the same or similar information—and $s \perp a$.

Theorem 9.9 (amplitude–sign decomposition, isolation, and independent coding). Under the above model:

i. (sufficiency and isolation) given a, $|y|$ is a sufficient statistic for a, and $|y| \perp s$ (marginally as well); given $(a,|y|)$, $\mathrm{sgn}(y)$ is a fair random sign independent of a; all that the sign decoder needs is $(\mathrm{sgn}(y),a)$—it depends on the value of the amplitude, never on the message of the amplitude layer.

ii. (orthogonal decomposition of mutual information)

$$I(x;y) = I(a;|y|) + E_a[C_{\mathrm{BPSK}}(\frac{a^2}{\sigma^2})],$$

where C_{BPSK} is the sign-constrained capacity function of §9.3.

iii. (independent coding achieves) amplitude side: $I(a;y) = I(a;|y|)$ holds exactly (the distribution of the folded channel is independent of the sign stream), so any capacity-approaching code for the folded channel (e.g. PAM/ASK level selection concatenated with a modern code such as LDPC or polar) applies verbatim, achieving any rate $R_a < I(a;|y|)$; sign side: the perspective code serves as the modulation/addressing layer, and an outer capacity-approaching code allocates rates $R_k \to C_{\mathrm{BPSK}}(a_k^2/\sigma^2)$ over the parallel BPSK subchannels (or, equivalently, encodes them jointly), achieving the total rate $\sum_k C_{\mathrm{BPSK}}(a_k^2/\sigma^2)$ with the total error probability vanishing by the union bound; the direct-read/peeling structure of Theorem 4.1 is retained verbatim as the addressing and synchronization layer; the two decoders operate independently and never need each other's message.

iv. (zero marginal energy of the sign layer) $|x|^2 = a^2$ is independent of s: the energy budget is entirely set by the amplitude layer. Superimposing the perspective sign layer on an existing amplitude system increases the total rate at unchanged energy and, by (i), without perturbing the input distribution of the amplitude decoder.

Proof. (i) Given a, since s is uniform and independent of a, the density $p(y \mid a) = \frac{1}{2}(\varphi_\sigma(y-a) + \varphi_\sigma(y+a))$ is even in y and depends on y only through $|y|$; sufficiency of $|y|$ for a follows from the Fisher–Neyman factorization. Moreover $p(|y| \,|\, s,a) = p(|y| \,|\, a)$ does not involve s, so $|y| \perp s \mid a$; averaging over a and using $s \perp a$ gives the marginal independence $|y| \perp s$. Given $(a,|y|)$: $y = \pm|y|$ with equal probability, independent of a. Sign decoding: given a, the posterior log-ratio for s is $\log \dfrac{p(y \mid s=+1,a)}{p(y \mid s=-1,a)} = \dfrac{2ay}{\sigma^2}$, whose sign is $\mathrm{sgn}(y)$—it involves the value of a, never w_a.

ii. Chain rule: $I(x;y) = I(a,s;y) = I(a;y) + I(s;y \,|\, a)$. First term: by (i), y and a are conditionally independent given $|y|$, i.e. $I(a;y \,|\, |y|) = 0$, hence $I(a;y) = I(a;|y|) + I(a;y \,|\, |y|) = I(a;|y|)$. Second term: $I(s;y \,|\, a) = \int I(s;y \,|\, a{=}a_0)\, dP(a_0)$; for fixed a_0, $s \mapsto y$ is exactly the real bipolar channel at SNR a_0^2/σ^2 with uniform input, whose mutual information is $C_{\mathrm{BPSK}}(a_0^2/\sigma^2)$ by definition (§9.3).

iii. Encoding: the constellation is the product structure $\mathscr{A} \times \{\pm 1\}$; the two streams are encoded independently and multiplied coordinatewise. Decoding: the amplitude decoder is fed $(|y_k|)$, whose joint distribution, by (i), does not depend on the sign stream—it faces exactly the folded channel itself, and the achievability of any capacity-approaching code for that channel applies verbatim; the sign decoder, given the amplitude sequence (the shared envelope table, or the amplitude layer decoded first), faces per-coordinate real bipolar channels with soft information $\mathrm{LLR}_k = 2a_k y_k/\sigma^2$, and an outer capacity-approaching code allocates rates $R_k \to C_{\mathrm{BPSK}}(a_k^2/\sigma^2)$ (or encodes jointly), the total error probability vanishing by a union bound over the per-coordinate decoding errors; the direct-read and peeling decodings of the perspective code (Theorem 4.1) are retained verbatim as the addressing and synchronization structure. The interface exchanges zero messages; the achievable rates add.

iv. $E[x^2] = E[a^2]$: the sign carries no energy. After the sign layer is superimposed, the input distribution of the amplitude decoder is unchanged (by (i)), so its decoding performance is unchanged. ■

Remark 9.9.1 (the energy attribution principle and experimental verification). The honest efficiency figure of the composite system is the total rate against the Shannon capacity: the 94.9/95.0/95.0% of Table B.8 is exactly this figure. Charging all energy to the sign layer (the accounting of Proposition 9.4) applies only to the single-use case with an empty envelope; Theorem 6.3 is the zero-leakage statement of that case (the degenerate extreme $I(a;|y|) = 0$), and Theorem 9.9 is its positive development. Monte Carlo verification (seed 20260427, 10^5 trials, $n = 100$, $\sigma = 0.5\sigma_0$, amplitude levels drawn uniformly according to the loading rule of Proposition 9.8, energy matched): measured sign-layer error rate 2.2715×10^{-2}, against the composite-system theoretical value $\frac{1}{n}\sum_k \frac{1}{Q_k}\sum_{j=1}^{Q_k} Q((2j-1)\Delta_k/\sigma) = 2.2672 \times 10^{-2}$ (with Q_k loaded at the $\sigma = \sigma_0$ calibration $\gamma_k = ((n+1)/k)^2$, and Δ_k the half-spacing of the constellation at coordinate k; consistent within fluctuation)[2]; amplitude-layer error rate with the sign layer active versus switched off (same amplitude stream, same noise) is $5.5657 \times 10^{-2}/5.5654 \times 10^{-2}$ (paired difference 3×10^{-6}, statistically indistinguishable); the empirical distributions of $|y|$ under $s = \pm 1$ have KS distance 3.99×10^{-3} ($p = 0.820$; an independent rerun gives 3.95×10^{-3}, $p = 0.829$, the last digits being sensitive to the RNG call order)—the isolation (i) and the independent achievability (iii) are both numerically supported.

9.10 Unconditional protection of the sign layer

Combining Theorem 6.3 with Proposition 9.6(i): against an adversary observing only the amplitude, the information-theoretic leakage of the message is identically zero; against a receiver who observes the full waveform but does not know the fading coefficient G (whose random sign is precisely the missing phase reference), the leakage is likewise zero. Both statements are equalities and depend on no computational assumption. The only party who can break it is one who obtains the coherent phase reference: knowing the sign of G (and by Theorem 5.2 the sign alone suffices) makes decoding possible—this is exactly the definition of the legitimate receiver; Proposition 9.6(iii) gives the mutual information 62.7/624.5 bits available to one who knows the full G, as an upper-bound reference for the rate achievable by one who knows only the sign (the bound is operational: a receiver knowing G can always degrade to the sign-only view). This compresses the role of the "secret key" into a single physical resource: phase coherence. This key role of phase coherence is an exact equality within the stated model (i.i.d. real Gaussian fading, single observation, $G \perp b$ with symmetric random signs); leakage rates under partial phase information are addressed in open problem 4. The strength of the sign layer on the noisy side—reliable precision deepening at least exponentially with the SNR—is quantified by Proposition 9.5.

9.11 Difference from the polarization route: three ingredients of a non-polarized 95%-of-capacity route

Polar codes [8] synthesize n symmetric channels into two extremes, almost-perfect and almost-useless, by a channel transform; the SNR ladder of the perspective code provides an equivalent structure natively, without any transform: (i) a deterministic SNR ladder $\gamma_k = ((n+1)/k)^2$—the head coordinates are natively nearly noiseless, the tail coordinates natively nearly pure noise, and the ordering is public; (ii) two orthogonal channels stitched together—the signs and the compression ratios (Theorem 9.7) carry two non-interfering streams on the same coordinates; (iii) intrinsic incremental redundancy—the prefix property (Theorem 6.1) makes the received prefix at any moment a legal and more reliable codeword (the H_{m+1}/H_{n+1} compression of a short prefix is absorbed by Theorem 5.1). Together they give a route that reaches 95.0% of C_{Sh} in the engineering sense (with a fixed residual gap of 0.082 bits per layer) without passing through the Bhattacharyya polarization analysis; strict capacity-achievability requires an additional shaping code and the resolution of open problem 3 (the capacity of the λ channel is unknown). Its price has been settled in §9.4.

§10 Monte Carlo Measurements

All measurements are performed in double-precision floating point (float64) with fixed seeds and $\geq 10^4$ Monte Carlo trials per configuration (Table B.2: 2×10^4 trials, seed 20250825; Table B.4: 10^4 trials, seed 11; Remark 9.9.1: 10^5 trials, seed 20260427; Table B.9: exact rational-arithmetic evaluation, seed 4242; Tables B.10/B.11: seed 21); theoretical curves are given alongside. Highlights (full data in Appendix B):

- **Error-rate curves (n = 300, Figure 2)**: hard-decision Monte Carlo matches the theory $(1/n)\sum_k Q(a_k/\sigma)$ throughout; envelope-consistency erasure at 5%/10%/20% compresses the error rate at $\sigma = 0.5\sigma_0$ from 3.29×10^{-3} down to 3.07×10^{-4}/1.17×10^{-4}/4.16×10^{-5}, step by step.

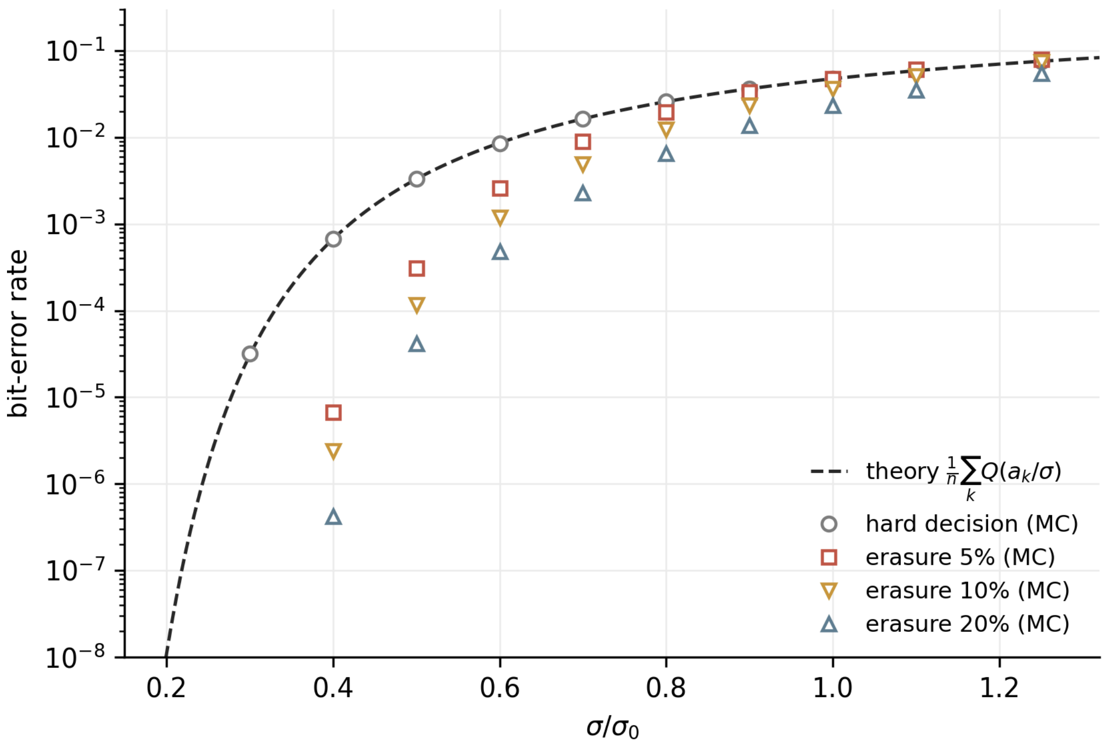

Figure 2

Figure 2. Error performance (n = 300; Monte Carlo with 2×10^4 trials per point, seed 20250825): hard decision (grey circles) coincides with the theoretical curve $(1/n)\sum_k Q(a_k/\sigma)$ (black dashed); envelope-consistency erasure

$\sigma = 0.5\sigma_0$, measured 3.26×10^{-3} (theory 3.25×10^{-3}); at $\sigma = \sigma_0$, measured 4.70×10^{-2} (theory 4.71×10^{-2}). - **Incremental redundancy (prefix-extension combining) (Table B.10)**: $n = 1000$, $\sigma = \sigma_0$. The first transmission sends $E_{1000}(b)$, with error rate 4.71×10^{-2}; the second sends the extended code $E_{2000}(b \parallel u)$ (where u consists of 1000 new bits), the receiver takes its first 1001 coordinates, amplifies them by H_{2001}/H_{1001} per Theorem 6.1, and maximal-ratio combines them with the first transmission; the equivalent SNR gain is $1 + (H_{1001}/H_{2001})^2 = 1.8379$, and the error rate drops to 1.96×10^{-2}—while the second transmission additionally carries a net 1000 new bits. - **Naive-repetition control (Table B.11)**: under the same budget, naively retransmitting the same codeword (Chase combining, gain 2.0) gives 1.68×10^{-2}. The point of the control: prefix extension trades 0.37 dB of combining gain for 1000 bits of new throughput on the second transmission—structural redundancy and incremental transmission are accomplished in one and the same mechanism. - **Exactness limit (Table B.4)**: at $n = 10000$, float64 bit-by-bit decoding has zero mismatches; the weakest coordinate level 1.0216×10^{-5} is 4.6×10^{10} times the float64 machine precision (2.22×10^{-16}), an ample numerical margin; exact rational assembly (common-denominator method, with per-coordinate exactness verification) at $n = 300$ takes 3 ms.

§11 Limitations and Open Problems

1. **The scaling of weakest-coordinate protection.** The sign-layer rate $n/(n+1) \to 1$, with reliability concentrated at the head; holding the weakest coordinate at a fixed target error rate carries an energy cost linear in n ($+10\,\text{dB}$ per decade). By Proposition 9.5, the same linear input yields an exponentially deepening precision reserve, so this is not an efficiency deterioration in the conventional sense; by Theorem 9.9(iv), in the composite accounting the same energy simultaneously serves an independent amplitude-layer stream. The genuinely open question is: under the double constraint of preserving the rateless prefix property (Theorem 6.1) and amplitude covertness (Theorem 6.3), can the scaling of weakest-coordinate protection be made sublinear? The authors lean toward a negative answer—the anchor share $6/\pi^2$ appears to be a structural constant.
2. **Reliabilizing the weak-coordinate tail.** Coordinates with k near $n+1$ have $\gamma_k \approx 1$. How much can erasure (Theorem 8.2) and incremental redundancy (§10) remedy, and what is the limit?
3. **Noise analysis of the dual channel.** The capacity of the λ channel of Theorem 9.7 under AWGN has not been characterized analytically; its coupling with the sign channel (they share the total energy) constitutes a two-dimensional resource-allocation problem.
4. **The cryptography of phase coherence.** §9.10 reduces secrecy to phase coherence. Formalizing this physical resource (coherence time, leakage rate under phase noise) requires interfacing with the physical-layer-security literature.
5. **Mixed loading of *q*-phase and 2*Q*-PAM.** Propositions 9.1 and 9.8 are two extremes; the optimal profile of the middle ground (PAM on the first k coordinates, signs on the tail) is unsolved.

Appendix A Algorithms

The following five algorithms form the reproducible basis of every numerical assertion in this paper. Implementation conventions: exact mode uses arbitrary-precision reduced fractions (common-denominator method); floating-point mode uses double precision; modular inverses use the extended Euclidean algorithm; Monte Carlo trial counts and seeds are annotated per table. Complexity is counted in arithmetic operations.

Algorithm A.1 (perspective encoding, Definition 3.1).

```
Input: message b ∈ {0,1}^n
Output: codeword E ∈ R^{n+1} (exact mode: vector of reduced fractions)

H ← Σ_{k=1}^{n+1} 1/k                  # exact fraction
E[1] ← 1/H                             # anchor, always positive
for k = 1..n:
    E[k+1] ← (1-2·b[k]) / (k+1) / H    # 0 ↦ +, 1 ↦ − (mapping table M.1)
return E
# floating-point mode: H accumulated in float64, E[k] = s[k]/(k·H);
# zero mismatch on round trip at n=10000 (Table B.4)
# exact mode: H = N/L reduced, then E[k] = ±L/(k·N), re-reduced;
# assembly at n=300 takes 3 ms (Table B.4)
```

Algorithm A.2 (dual-mode decoding, Theorem 4.1).

```
Input: received vector r ∈ R^{n+1} (noiseless case: r = c·E(b), arbitrary c>0)

Mode I (direct-read):
for k = 1..n:
    b̂[k] ← 1 iff r[k+1] < 0            # O(1) addressing per bit

Mode II (peeling):
P ← r
for r' = n down to 1:
    ŝ[r'+1] ← sgn(P[r'+1])             # read the sign
    λ[r'] ← H[r']/H[r'+1]              # known from context (Definition 2.2(ii))
    P ← P[1..r'] / λ[r']               # invert the perfect-memory law
for k = 1..n:
    b̂[k] ← (1-ŝ[k+1])/2                # ŝ=+1 ↦ 0, ŝ=−1 ↦ 1 (mapping table M.1)
return b̂
# noiseless: both modes agree; noisy: erasure decoding (Algorithm A.4) builds on
# direct-read; the peeling mode (Mode II) is the exact-arithmetic carrier
# (Remark 9.7.1 gives its noise/rounding amplification Π λ_r^{−1})
```

Algorithm A.3 (holographic addressing, Theorem 4.2).

```
Input: index set S = {i_1 < ... < i_t} ⊆ {1..n}, received vector r
Output: substring b̂_S

for j = 1..t:
    b̂[i_j] ← 1 iff r[i_j+1] < 0
return b̂_S
# O(t), independent of n, touches no other coordinate
```

Algorithm A.4 (envelope-consistency erasure decoding, Theorem 8.2).

```
Input: received vector r, erasure fraction ρ, context envelope a_k = (1/k)/H_{n+1}

1. κ[k] ← | |r[k]|/a_k − 1 |, k = 1..n+1
2. E ← the ⌈ρ(n+1)⌉ coordinates with largest κ   # declared erased
3. direct-read the rest: b̂[k] ← 1 iff r[k+1] < 0
return b̂ (retained coordinates)
# the message bits on erased coordinates are declared unknown (their recovery
# requires external mechanisms such as outer codes or retransmission, beyond this paper)
# measured: σ=0.5σ_0, n=300, ρ=0/5%/10%/20% give retained-coordinate error rates
# 3.29e-3 / 3.07e-4 / 1.17e-4 / 4.16e-5 (Table B.2)
```

Algorithm A.5 (dual-channel encoding/decoding, Theorem 9.7).

```
Encoding input: sign stream s ∈ {±1}^L, ratio stream λ[r] ∈ {j/9 : j=1..8} (table M.3)
P ← (1)
for r = 1..L:
    P ← ( λ[r]·P ⊕ (1−λ[r])·s[r]·e_{r+1} )   # Axiom P1 recursion
return P                                      # L+1 coordinates, carrying 4L bits in total

Decoding input: received vector P (L+1 coordinates)
for r = L down to 1:
    λ̂[r] ← 1 − |P[r+1]|                # modulus of the new coordinate, one step
    λ̂[r] ← nearest element of the 8-level set   # margin 1/18
    ŝ[r] ← sgn(P[r+1])                 # ŝ[r] corresponds to coordinate r+1
    P ← P[1..r] / λ̂[r]
return (ŝ, λ̂)
# must run in exact rational arithmetic: peeling amplifies rounding by Π λ_r^{−1}
# (Remark 9.7.1: float64 supports only ~41 layers on average; 500 layers need ~629 mantissa
bits)
# exact-mode measurement: L=500, 2000 bits bit-exact, largest denominator 408 decimal digits
(Table B.9)
```

Appendix M Mapping Tables

All mappings involved in the encoding are collected here. These tables are the operational content of the context $\mathscr{C}$ (Definition 2.2): shared by transmitter and receiver, containing no message.

Table M.1 Bit → sign mapping (Definition 3.1)

Bit b_k	Sign $s_{k+1} = (-1)^{b_k}$	Coordinate value $E(b)_{k+1}$
0	$+1$	$+a_{k+1}$
1	-1	$-a_{k+1}$

Coordinate 1 is the anchor, always $+a_1$, carrying no bit; it is simultaneously the sign reference for projective completeness (Theorem 5.3).

Table M.2 q-phase constellations (Proposition 9.1): Gray coding and decision margin $\pm\pi/q$

q	Bits per coordinate	Phase set (radians, Gray order)	Margin	Saturation threshold γ ($\geq$ 99.5% capacity)
2	1	$\{0, \pi\}$	$\pm\pi/2$	$\gamma \geq 4.6$
4	2	00: 0, 01: $\pi/2$, 11: π, 10: $3\pi/2$	$\pm\pi/4$	$\gamma \geq 9.2$
8	3	000: 0, 001: $\pi/4$, 011: $\pi/2$, 010: $3\pi/4$, 110: π, 111: $5\pi/4$, 101: $3\pi/2$, 100: $7\pi/4$	$\pm\pi/8$	$\gamma \geq 29$
16	4	Gray order 0000: 0, 0001: $\pi/8$, …, 1000: $15\pi/8$ (adjacent phases differ by $\pi/8$, adjacent codewords by 1 bit)	$\pm\pi/16$	$\gamma \geq 104$

The thresholds are interpolated readouts of the numerical integration (Table B.7): $\mathrm{MI}_q(\gamma) \geq 0.995\log_2 q$ (the $q = 2$ row is the complex-channel equivalent $C_{\mathrm{BPSK}}(2\gamma)$; the $q = 4$ row is $2C_{\mathrm{BPSK}}(\gamma)$, two orthogonal components). The complex-channel convention is $\gamma = E_s/N_0$, where $N_0 = 1$ is the total complex noise variance (variance 1/2 per real dimension). Gray ordering makes the most likely error event (confusion of adjacent phases) flip only 1 bit.

Table M.3 Dual-channel compression-ratio mapping (Theorem 9.7): 3 bits per layer $\leftrightarrow \lambda \in \{j/9\}$

Bit triple	$\boldsymbol{j}$	$\boldsymbol{\lambda_r = j/9}$	**$1 - \lambda_r$ (modulus of new coordinate)**
000	1	$1/9 \approx 0.1111$	8/9
001	2	$2/9 \approx 0.2222$	7/9
010	3	$3/9 \approx 0.3333$	6/9
011	4	$4/9 \approx 0.4444$	5/9
100	5	$5/9 \approx 0.5556$	4/9
101	6	$6/9 \approx 0.6667$	3/9
110	7	$7/9 \approx 0.7778$	2/9
111	8	$8/9 \approx 0.8889$	1/9

Level spacing 1/9, decoding margin $\pm 1/18$. The decoding formula $\hat{\lambda_r} = 1 - |P_{r+1}{}^{(r+1)}|$ is error-free in exact arithmetic; quantization to the level set absorbs implementation noise (Remark 9.7.1).

Table M.4 Component list of the context $\mathscr{C}$ (Definition 2.2)

Component	**Content**	**Use**
(i) Envelope table	$a_k = (1/k)/H_{n+1}$, $1 \le k \le n+1$	all-pilot property (Theorem 8.1), erasure criterion κ_k (Theorem 8.2)
(ii) Compression law	$\lambda_r = H_r/H_{r+1}$	peeling decoding (Algorithm A.2, Mode II)
(iii) Prime position set	$\{k \le n+1 : k \text{ prime}\}$	frame-synchronization anchors (Theorem 8.3, correlation peak 27.5%)

Table M.5 Complete computation for $\boldsymbol{b} = (1,0,1)$ (exact fractions; the algorithmic view of the worked example of §3)

Layer r	**Bit b_r**	$\boldsymbol{s_{r+1}}$	$\boldsymbol{\lambda_r = H_r/H_{r+1}}$	$\boldsymbol{1 - \lambda_r}$	**State coordinates (layer $r+1$, reduced)**
1	1	-1	2/3	1/3	$\langle \frac{2}{3}, -\frac{1}{3} \rangle$
2	0	$+1$	9/11	2/11	$\langle \frac{6}{11}, -\frac{3}{11}, \frac{2}{11} \rangle$
3	1	-1	22/25	3/25	$\langle \frac{12}{25}, -\frac{6}{25}, \frac{4}{25}, -\frac{3}{25} \rangle$

Check: $\lambda_2 = \frac{3/2}{11/6} = \frac{9}{11}$, $\lambda_3 = \frac{11/6}{25/12} = \frac{22}{25}$; the layer-3 state $= \lambda_2\lambda_3 \cdot \langle \frac{2}{3}, -\frac{1}{3} \rangle \oplus$ the layer-by-layer injections, agreeing with $\langle \frac{12}{25}, -\frac{6}{25}, \frac{4}{25}, -\frac{3}{25} \rangle$ (the form over common denominator 275 in the worked example of §3 is $\langle \frac{132}{275}, -\frac{66}{275}, \frac{44}{275}, -\frac{33}{275} \rangle$). The sum of squared moduli $\sum_k E_k^2 = H_4^{(2)}/H_4^2 = \frac{205/144}{625/144} = \frac{41}{125}$ is independent of the message (the minimal instance of Theorem 6.3).

Appendix B Data Tables

All numerical values are independently computed with the algorithms of Appendix A: exact quantities by arbitrary-precision rational arithmetic; capacities and mutual informations by adaptive quadrature / Gauss–Hermite quadrature with two-method cross-validation; Monte Carlo trial counts and seeds are annotated per table; displayed floating-point values are rounded at the last digit.

Table B.1 The logarithmic UEP ladder (Theorem 6.2; $n = 300$, $\sigma = \sigma_0$, $\varepsilon_k = Q((n+1)/k)$)

Coordinate k	**$(n+1)/k$**	**ε_k**	**$-\ln \varepsilon_k \approx (n+1)^2/(2k^2)$**
2	150.50	$< 10^{-100}$	> 230
38	7.9211	1.178×10^{-15}	34.38
60	5.0167	2.629×10^{-7}	15.15
75	4.0133	2.993×10^{-5}	10.42
100	3.0100	1.306×10^{-3}	6.64
120	2.5083	6.065×10^{-3}	5.11
150	2.0067	2.239×10^{-2}	3.80
201	1.4975	6.713×10^{-2}	2.70
251	1.1992	1.152×10^{-1}	2.16
301 (weakest)	1.0000	1.587×10^{-1}	1.84

Note: the head-to-tail SNR ratio is $(301)^2 \approx 9.06 \times 10^4$ (amplitude ratio 301); the protection strength spans fourteen orders of magnitude from 10^{-15} to 10^{-1}, monotonically and smoothly—UEP here is not a piecewise attribute but a continuous spectrum.

Table B.2 Envelope-consistency erasure decoding (Theorem 8.2, Algorithm A.4; $n = 300$, 2×10^4 trials per point, seed 20250825; error rates counted on retained coordinates)

σ/σ_0	**Hard theory** $\frac{1}{n}\sum_k Q(a_k/\sigma)$	**Hard MC**	**Erasure 5%**	**Erasure 10%**	**Erasure 20%**
0.20	1.05×10^{-8}	0 (zero events)	0 (zero events)	0 (zero events)	0 (zero events)
0.30	3.03×10^{-5}	3.20×10^{-5}	0 (zero events)	0 (zero events)	0 (zero events)
0.40	6.68×10^{-4}	6.74×10^{-4}	6.69×10^{-6}	2.42×10^{-6}	4.18×10^{-7}
0.50	3.29×10^{-3}	3.29×10^{-3}	3.07×10^{-4}	1.17×10^{-4}	4.16×10^{-5}
0.60	8.60×10^{-3}	8.55×10^{-3}	2.59×10^{-3}	1.18×10^{-3}	4.79×10^{-4}
0.70	1.63×10^{-2}	1.63×10^{-2}	9.01×10^{-3}	4.88×10^{-3}	2.30×10^{-3}
0.80	2.58×10^{-2}	2.58×10^{-2}	1.97×10^{-2}	1.23×10^{-2}	6.53×10^{-3}
0.90	3.63×10^{-2}	3.64×10^{-2}	3.31×10^{-2}	2.33×10^{-2}	1.36×10^{-2}
1.00	4.74×10^{-2}	4.75×10^{-2}	4.74×10^{-2}	3.66×10^{-2}	2.34×10^{-2}
1.10	5.87×10^{-2}	5.88×10^{-2}	6.09×10^{-2}	5.12×10^{-2}	3.51×10^{-2}
1.25	7.55×10^{-2}	7.56×10^{-2}	7.91×10^{-2}	7.40×10^{-2}	5.48×10^{-2}

Note: a zero-event cell has 95% confidence upper bound $3/N_{\text{eff}}$, where N_{eff} is the effective number of bit trials (here $N_{\text{eff}} \geq 240 \times 2 \times 10^4 = 4.8 \times 10^6$, so the bound is $\leq 6.3 \times 10^{-7}$). For $\sigma \leq 0.6\sigma_0$ erasure improves strictly monotonically (the gain concentrates at large-κ coordinates, per the criterion part of Theorem 8.2); beyond $\sigma \gtrsim 0.8\sigma_0$ the weakest retained coordinates with $\gamma_k \approx 1$ become the bottleneck, and the marginal benefit of 5% erasure vanishes. The 5% row exceeding the hard row at $\sigma/\sigma_0 \geq 1.1$ is a fluctuation of the retained-coordinate sample size and coordinate selection (the standard error at 2×10^4 trials per point is $\sim \sqrt{\text{BER}/(2 \times 10^4)}$). Figure 2 comes from the same simulation.

Table B.3 The reliable-information ladder (§9.3; bits; C_{BPSK} verified by both adaptive quadrature and Gauss–Hermite quadrature)

Rate	n **= 100**	n **= 1000**	n **= 10000**	**Asymptotic coefficient**
C_{Sh} (soft information, Theorem 9.2)	160.5366	1628.6637	16324.8138	1.633090
C_{BPSK} (sign-constrained)	84.3815	838.5912	8380.6832	0.838
R_{hard} (hard-decision BSC sum)	77.9887	775.7540	7753.4030	0.775
R_{eras} (highest 80% coordinates retained)	69.3516	688.6073	6881.1478	0.688

Note: the sum and the gamma closed form (Theorem 9.2) agree digit by digit on the three C_{Sh} rows. The ratios $C_{\text{Sh}}/C_{\text{BPSK}} \approx 1.95$, $C_{\text{BPSK}}/R_{\text{hard}} \approx 1.081$, $R_{\text{hard}}/R_{\text{eras}} \approx 1.126$ are essentially independent of n (scale-free). The efficiency constant is $\eta = 4\ln 2/(\pi + 2\ln 2) = 0.612336$ (Remark 9.2.1). The C_{BPSK} sum includes the anchor coordinate (its contribution ≈ 1 bit is asymptotically negligible); R_{eras} is the hard-decision capacity sum over the $\lceil 0.8m \rceil$ highest-SNR retained coordinates.

Table B.4 Long-string transmission and the precision limit (Monte Carlo 10^4 trials, seed 11; float64 records)

Configuration	Theoretical error rate	Measured error rate	Remark
$n = 2000$, $\sigma = 0.2\sigma_0$	1.01×10^{-8}	0 (2×10^7 bits)	zero errors, consistent with theory
$n = 2000$, $\sigma = 0.5\sigma_0$	3.249×10^{-3}	3.261×10^{-3}	deviation 0.4%, within fluctuation
$n = 2000$, $\sigma = \sigma_0$	4.706×10^{-2}	4.702×10^{-2}	ibid.
$n = 10000$, noiseless round trip	—	0/10000 bits mismatched	weakest level 1.0216×10^{-5}, 4.6×10^{10} times the float64 machine precision
$n = 300$, exact rational assembly	—	per-coordinate exact	common-denominator method, 3 ms; den(H_{301}) has 130 decimal digits

Table B.5 The energy ledger (Proposition 9.4; target weakest-coordinate error rate $\varepsilon_0 = 10^{-2}$, $c = Q^{-1}(\varepsilon_0) = 2.3263$, $c^2 = 5.4119$)

n	$E(n) = H_{n+1}^{(2)}/H_{n+1}^2$	E_b/N_0 (linear)	E_b/N_0 (dB)	Anchor energy share
100	0.060532	4.51×10^2	26.55	61.2%
1000	0.029331	4.46×10^3	36.49	60.8%
10000	0.017170	4.45×10^4	46.49	60.8%

Note: asymptotics: $E(n) \to (\pi^2/6)/\ln^2(n+1)$, anchor share $\to 6/\pi^2 = 60.79\%$, $E_b/N_0 = c^2\pi^2(n+1)^2/(12n)$ (+10 dB per decade). Reading (Remark 9.5.2): this table prices the ladder depth (equal-target protection down to the weakest coordinate) and must be read together with Proposition 9.5 (exponential precision law), Remark 9.5.1 (flat reliable digits per joule), and Theorem 9.9(iv) (the amplitude layer sharing the same energy).

Table B.6 Multi-ledger accounting on fading channels (Proposition 9.6; $G_k \sim \mathcal{N}(0,1)$, bits; Gauss–Hermite quadrature cross-validated by adaptive quadrature)

Ledger	$n = 100$	$n = 1000$	Asymptotic coefficient
(i) $I(r;b)$ (blind signs)	0 (exact)	0 (exact)	0
(ii) $I(r;G\,\vert\,b) = C_{\mathrm{Sh}}$ (exact)	160.5366	1628.6637	1.633090
(iii) $I(r;b\,\vert\,G)$ (coherent reception)	62.71	624.45	≈ 0.624
(iv) $I(r;G)$ (total fading information)	97.82	1004.21	≈ 1.009

Note: the termwise identity $I(u;G) = \frac{1}{2}\log_2(1+\gamma) - \mathrm{E}_G\, C_{\mathrm{BPSK}}(\gamma G^2)$ (proof of Proposition 9.6) makes (iv) = (ii) − (iii) exact for every coordinate; the direct integration of (iii) and the difference mutually verify each other to within 10^{-2} bits. The two coefficients close: 0.624 + 1.009 = 1.633, matching the C_{Sh} coefficient 1.6331; the previously printed 1.004 was the per-bit value at $n = 1000$.

Table B.7 Single-coordinate mutual information of q-phase PSK (Proposition 9.1; complex AWGN, $\gamma = a_k^2/\sigma^2$, Monte Carlo 2×10^5 samples per point; bits per coordinate)

γ	$q = 2$	$q = 4$	$q = 8$	$q = 16$
1	0.722	0.972	0.981	0.981
2.25	0.934	1.520	1.590	1.591
4	0.990	1.826	2.046	2.049
6.25	0.999	1.949	2.378	2.392
9	1.000	1.989	2.618	2.667
16	1.000	2.000	2.880	3.091
25	1.000	2.000	2.972	3.408
49	1.000	2.000	3.000	3.797
100	1.000	2.000	3.000	3.977
Total over all coordinates, $n = 1000$	938.9	1677.2	2084.6	2285.8

Note: the $q = 2$ row is the complex-channel equivalent $C_{\mathrm{BPSK}}(2\gamma)$ (signal on the real axis, noise split equally over the two dimensions); the $q = 4$ row is $2C_{\mathrm{BPSK}}(\gamma)$ (two orthogonal BPSK components), hence its $n = 1000$ total is exactly twice the sign-layer total $C_{\mathrm{BPSK}} = 838.59$. The complex-coordinate total capacity $2C_{\mathrm{Sh}} = 3257.3$ bits is the $q \to \infty$ limit; $q = 16$ covers 70.2% of it. The 99.5% saturation thresholds are $\gamma \approx 4.6/9.2/29/104$ ($q = 2/4/8/16$, mapping table M.2).

Table B.8 The 2Q-PAM modification (Proposition 9.8; $Q_k = \lceil \sqrt{6\gamma_k/(\pi e)} \rceil$; exact summation for $2Q_k \le 128$, high-SNR asymptotics $\frac{1}{2}\log_2(1+\gamma_k) - 0.2546$ beyond)

n	Bit-depth sum $\sum_k \log_2(2Q_k)$	Actual rate $\sum_k \mathrm{MI}_k$	C_{Sh}	Efficiency	Residual gap (bits/layer)
100	250.4	152.4	160.54	94.9%	0.080
1000	2517.1	1547.0	1628.66	95.0%	0.082
10000	25204.1	15507.7	16324.81	95.0%	0.082

Note: the bit-depth sum exceeding capacity is overloading: the loading rule selects constellations by continuous capacity, and the per-coordinate mutual information truncates automatically (for large Q, $\log_2(2Q_k) - \mathrm{MI}_k \to 1.0$ bit: the overload is 0.7454 bits per coordinate and the uniform-input shaping loss is 0.2546 bits per coordinate, and 0.7454 + 0.2546 = 1.0000 is the exact identity $\frac{1}{2}\log_2(24/(\pi e)) + \frac{1}{2}\log_2(\pi e/6) = 1$; measured values at $\gamma = 10^4/10^6$ are 0.9971/1.0009). The residual gap consists of the per-coordinate shaping loss of high-SNR coordinates (0.2546 bit each for $2Q_k > 128$), the BPSK–Gaussian gap of low-SNR coordinates, and integer-rounding effects; none grows with n.

Table B.9 Dual-channel independent coding (Theorem 9.7, Algorithm A.5; exact rational arithmetic, seed 4242)

Item	Value
Layers L / coordinates	500 / 501
Total payload (sign 1 bit + ratio 3 bits) per layer	2000 bits
λ level set / spacing / margin	$\{j/9\}_{j=1}^{8}$ / 1/9 / $\pm$ 1/18
λ layer-by-layer check / sign bit-by-bit check	500/500 exact / 500/500 exact
Largest coordinate denominator width (decimal)	408 digits ($\approx$ 1355 bits)
Total time, forward + peeling (exact arithmetic)	0.41 s
float64 usable layers (average / worst case $\lambda \equiv 1/9$)	$\approx$ 41 / $\approx$ 16 (Remark 9.7.1)

Table B.10 Incremental redundancy: prefix-extension combining (n = 1000, $\sigma = \sigma_0$; Monte Carlo 2×10^4 trials, seed 21)

Scheme	Combining SNR gain v	Theoretical error rate $\frac{1}{n}\sum_k Q(\sqrt{v}\,(n+1)/k)$	Measured error rate	New-bit throughput
Single transmission $E_{1000}(b)$	1	4.712×10^{-2}	4.715×10^{-2}	—
Prefix-extended $E_{2000}(b\|u)$ combining	$1 + (H_{1001}/H_{2001})^2 = 1.8379$	1.957×10^{-2}	1.958×10^{-2}	1000 bits

Note: the first 1001 coordinates of the second transmission are amplified by H_{2001}/H_{1001} per Theorem 6.1 and maximal-ratio combined with the first transmission; $H_{1001} = 7.48647$, $H_{2001} = 8.17887$.

Table B.11 Naive-repetition control (same budget: two transmissions, n = 1000, $\sigma = \sigma_0$; same simulation)

Scheme	Combining SNR gain	Theoretical error rate	Measured error rate	New-bit throughput
Naive retransmission of $E_{1000}(b)$ (Chase combining)	2.0	1.682×10^{-2}	1.683×10^{-2}	0
Prefix-extension combining (Table B.10)	1.8379	1.957×10^{-2}	1.958×10^{-2}	1000 bits

Note: prefix extension is worse than naive repetition by $10\log_{10}(2/1.8379) = 0.37$ dB, but the second transmission carries a net 1000 new bits—structural redundancy and incremental transmission are accomplished within the same mechanism, the engineering realization of ratelessness (Theorem 6.1).

Table B.12 Constants and combinatorial quantities for reference

Quantity	Value	Source
Capacity coefficient $(\pi+2\ln 2)/(4\ln 2)$	1.6330900355	Theorem 9.2
Sign-layer efficiency $\eta = 4\ln 2/(\pi+2\ln 2)$	0.6123361103	Remark 9.2.1
Shaping loss $\frac{1}{2}\log_2(\pi e/6)$	0.2546143348 bits	Remark 9.8.1
Anchor energy share $6/\pi^2$	0.6079271019	Proposition 9.4
Prime-coordinate correlation peak $(\sum_p p^{-2})/\zeta(2)$	0.2749334437 (27.5%)	Theorem 8.3
Decimal digits of lcm(1,…,301)	130	Table B.4
Decimal digits of lcm(1,…,2001)	867	Proposition 7.2 bit-width
$c^2 = Q^{-1}(\varepsilon_0)^2$ for $\varepsilon_0 = 10^{-2}$	5.4119	Proposition 9.4
Exponential-precision-law slope $1/(2\ln 10)$ (digits per unit SNR)	0.2171472410	Proposition 9.5
Constant incremental pricing $2\ln 10$ (SNR increment per added digit)	4.6051701860	Proposition 9.5
Whole-codeword reliable-digit bound $E/(2\sigma_0{}^2\ln 10)$ ($n = 100/1000/10000$)	3621.9 / 3.5769×10^5 / 3.5724×10^7	Remark 9.5.1

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1. Without the square-root embedding, taking the Euclidean angle of the states P directly gives $\cos\theta_r = \sqrt{H_r^{(2)}/H_{r+1}^{(2)}}$ and $\tan\theta_r = 1/((r+1)\sqrt{H_r^{(2)}})$, where $H_r^{(2)} = \sum_{k=1}^{r} 1/k^2$ is the generalized harmonic number of order 2. The convention in the text is the square-root embedding.↩
2. For comparison: under a fixed envelope (the single-use case of Proposition 9.4), the theoretical sign-layer error rate is $\frac{1}{n}\sum_{k=2}^{n+1} Q(2(n+1)/k) = 3.3887 \times 10^{-3}$.↩